

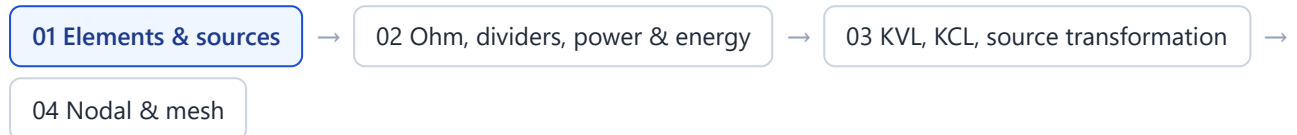
Circuits & Systems — Study Pack (MTE)

ECE2107 · B.Tech ECE Sem 3 · follows the professor's MTE syllabus (dated 10/9/26) · built from two sets of class notes · every answer checked by circuit simulation

The path through this pack

The order is the professor's MTE syllabus, item by item.

Part I — elements, laws, analysis



Part II — theorems (DC and AC)



Part III — transient analysis



Each file needs the ones before it. 04 is KCL/KVL (03) written as a method. 06 uses 05's "switch a source off" rule, 04's nodal analysis and 01's dependent sources. 07 is built on 06. 08 uses 06 for the time constant; 09 uses 08's Laplace route.

File	Syllabus item	What it teaches	Board Qs
01	1 · elements (R, L, C, IDS, DS)	active vs passive, independent and dependent sources (with devices), R, L, C laws, what can't jump, lumped vs distributed	—
02	2 · laws	Ohm, voltage and current division (DC and AC), power sign convention, energy, ladders	4
03	3 · KVL, KCL, source transformation	the two laws, power balance, source transformation	1 + 1 textbook
04	4 · nodal and mesh	both methods, with current sources and dependent sources	1 + 1 textbook
05	5 · superposition	method, the cases where it fails, AC sources at different frequencies	2 + 1 textbook
06	5 · Thévenin & Norton	with a dependent source; with AC impedances	1 + 1 textbook
07	5 · maximum power transfer	$R_L = R_{Th}$, efficiency, conjugate match (worked)	2 sets + 1 textbook
08	6 · first order	$0^- / 0^+ / \infty$, τ , the one formula, Laplace, test signals δ , u , r	4 + 2 textbook
09	7 · second order	series RLC: over-, critically, under-damped, undamped	textbook
10	all	one paper: 12 theory prompts + 20 numericals	14 + 6 textbook

Where the questions come from

A question that opens a section is **quoted from the professor's board** where one exists, and tagged with its notes page: **Class p.7** = first notes set (the 23-page scan), **Set 2 p.9** = second notes set (22 photos). Where the syllabus names a topic the class has not yet worked a question on, the pack uses a standard textbook-style problem, tagged **textbook**.

	Base	Status
P1	The professor's board questions (notes sets 1 and 2)	14 questions , all used
P2	Textbook problems	7 , written for syllabus items with no board question (KVL, nodal/mesh, AC theorems, Laplace, test signals, RLC). Values chosen so answers are clean; all checked by simulation

Marks per question: **not known** (no hand-out, no past paper yet).

How each file runs

Part	What happens
Map	the file's pieces, in order
Build	each step opens with a real question → you guess → the answer teaches the idea → a Check
Exam form	what to reproduce in the exam: statements, procedures, the formula sheet
Traps	the mistakes that cost marks
Self-test, Answers	answers are on the last page(s) of every file

THE ONE RULE THAT MAKES THIS WORK

When you reach a **Guess first** box, write a guess on paper **before** reading on: one line, however wrong. A wrong guess that gets corrected sticks better than a right answer you only read.

Mark every Check and Self-test item:

- **clean** → revisit in 3 days
- **with struggle** → revisit tomorrow
- **wrong** → redo today, after the file

Units and symbols used everywhere

- Uppercase **V, I** = DC (constant) values. Lowercase **v, i** = values that change with time. Complex values such as $5 - j5$ V are **phasors**: amplitude and phase in one number.
- Resistances in **kΩ** with currents in **mA** give voltages in **V** directly: $2 \text{ k}\Omega \times 3 \text{ mA} = 6 \text{ V}$. Most questions use this.
- **R_{Th}, V_{Th}** = Thévenin resistance and voltage. **I_N** = Norton current. **R_L** = load resistance.
- **0⁻** = just before the switch moves. **0⁺** = just after. **∞** = long after (steady state).

Where the two notes sets disagree with the pack

Topic	Notes say	Pack says	Why
Superposition, two 5 V sources (05 §5)	set 2: $0 + 0 = 0$ A, "LIMITATION"; set 1: 10 mA	0 A by superposition, true 5 mA	set 2 is the correct use of the rule and shows why it fails; the pack follows set 2
Thévenin with 2I (06 §2)	set 2: $V_{Th} = 5.0025$ V	20/3 V	set 2 reads 2I as $2 \Omega \times I$; the pack reads $2 \text{ k}\Omega \times I$. Both are right for their reading. Ask the professor
RL switch that opens (08 §5)	$I_R(0^+) = 0$, $I_L(0^+) = 10$ mA	cannot both hold in one series loop	ask the professor which path the current takes

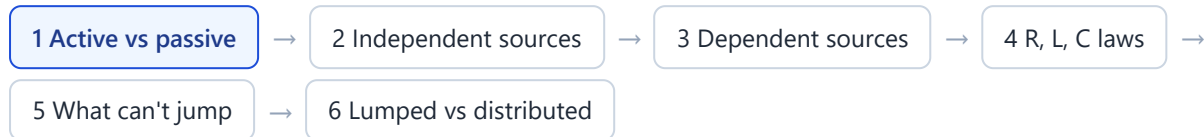
How this pack was checked

- Facts: checked against the textbook-sourced circuits knowledge base (Hayt, Van Valkenburg, Sudhakar & Shyammohan) and Alexander & Sadiku.
- Numbers: every circuit in this pack was solved a second time by computer (nodal analysis and time simulation), and the answers here match. The script is `src/verify_mte.py`.
- Where the class notes had a slip of the pen, the pack uses the corrected version and says so in a Note.

01 — Circuit elements and sources

Class notes p.1–3, set 2 p.1–3 · MTE syllabus item 1 · needed by every other file · no board question yet: the openers here are Predict prompts, not exam questions

Map



The professor's classification (class p.1):

Element	Kind	Examples
Active · independent	fixed value	voltage source, current source
Active · dependent	value set by another circuit quantity	VCVS, VCCS, CCVS, CCCS
Passive	cannot deliver net energy	R, L, C

Build

1 · Active vs passive

PREDICT

A phone charger's transistor amplifies a signal. Does the transistor create energy, or take it from somewhere?

An **active element can deliver net energy** to the rest of the circuit. A **passive element cannot**:

- R turns energy into heat.
- L and C store energy and give it back, but never more than they took.

An amplifier (transistor, op-amp) is **active**. It does not create energy: it draws it from its DC supply and uses it to make a bigger copy of the input signal.

WORDING IN THE CLASS NOTES

Class p.2 says an active element "doesn't require any external energy source". Read that as: *the element is the thing that supplies energy to the circuit*. It does not mean an amplifier runs without a supply. It doesn't.

CHECK 1

(a) Is a battery being charged acting as active or passive at that moment? (b) Why is an op-amp called active even though it needs ± 15 V to work?

2 · Independent sources

PREDICT

An ideal 10 V source is connected to 1 Ω , then to 1 M Ω . What happens to its voltage and to its current in each case?

- **Ideal voltage source:** keeps its voltage **fixed whatever current flows**. Internal resistance = **0**.
- **Ideal current source:** keeps its current **fixed whatever voltage appears**. Internal resistance = ∞ .

Those two internal resistances return in 05: "switching off" a source means replacing it with its internal resistance, so a V source becomes a **short** and an I source becomes an **open**.



Independent V
fixed V



Independent I
fixed I



VCVS
 $V = \mu V_x$



CCVS
 $V = r I_x$



VCCS
 $I = g V_x$



CCCS
 $I = \beta I_x$

Circle = independent source. Diamond = dependent source. + / - inside = voltage source; arrow inside = current source.

3 · Dependent (controlled) sources

PREDICT

A transistor's collector current is about 100 times its base current. What kind of source would you draw to model that?

A **dependent source** outputs a value set by a voltage or current **somewhere else in the circuit** (class p.2: "value depends on another parameter; it is not constant").

Name	Output	Controlled by	Law	Unit of the constant
VCVS	voltage	a voltage V_x	$V = \mu \cdot V_x$	none
CCVS	voltage	a current I_x	$V = r \cdot I_x$	Ω
VCCS	current	a voltage V_x	$I = g \cdot V_x$	S (siemens)
CCCS	current	a current I_x	$I = \beta \cdot I_x$	none

The class names them by what they **depend on** (set 2 p.2): VDVS = VCVS, VDVS = VCCS, CDVS = CCVS, CDVS = CCCS. Same four sources.

The class homework (set 2 p.2–3): find a real device for each.

Source	Device	Why it fits
VCVS	op-amp (IC 741)	output voltage = gain $\times$ input voltage
CCCS	BJT (BC547)	$I_C = \beta \cdot I_B$
VCCS	MOSFET	drain current is set by the gate voltage: $i_D = g_m \cdot V_{GS}$
CCVS	transimpedance amplifier (the class answer: optical fibre)	turns a photodiode's current into a voltage, e.g. in an optical-fibre receiver

UNITS OF A CCVS

In 06 the source is marked **2I**. A voltage equal to "2 $\times$ a current" only makes sense if 2 carries units of Ω . The pack reads it with $k\Omega$ and mA, so 2I means **2 $k\Omega \times I$** : $I = 1.5 \text{ mA}$ gives 3 V. The class working in set 2 reads it as **2 $\Omega \times I$** instead, and gets a different answer. 06 shows both.

CHECK 2

(a) Draw the symbol for a CCCS with value $3I_1$. (b) Which of the four is the op-amp model, and what is μ called for an op-amp?

4 • The R, L, C laws

PREDICT

If the current in an inductor is constant, what voltage appears across it?

	Resistor R	Inductor L	Capacitor C
Law	$v = iR$	$v = L \cdot \frac{di}{dt}$	$i = C \cdot \frac{dv}{dt}$
Where it comes from	Ohm: $V \propto I$	$\psi = N\phi = Li$, and $v = \frac{d\psi}{dt}$	$q = Cv$; differentiate: $\frac{dq}{dt} = i$
Energy	dissipated as heat	stored in the magnetic field: $\frac{1}{2}Li^2$	stored in the electric field: $\frac{1}{2}Cv^2$
DC steady state	resistor	short circuit ($di/dt = 0 \Rightarrow v = 0$)	open circuit ($dv/dt = 0 \Rightarrow i = 0$)

SLIP IN THE CLASS NOTES

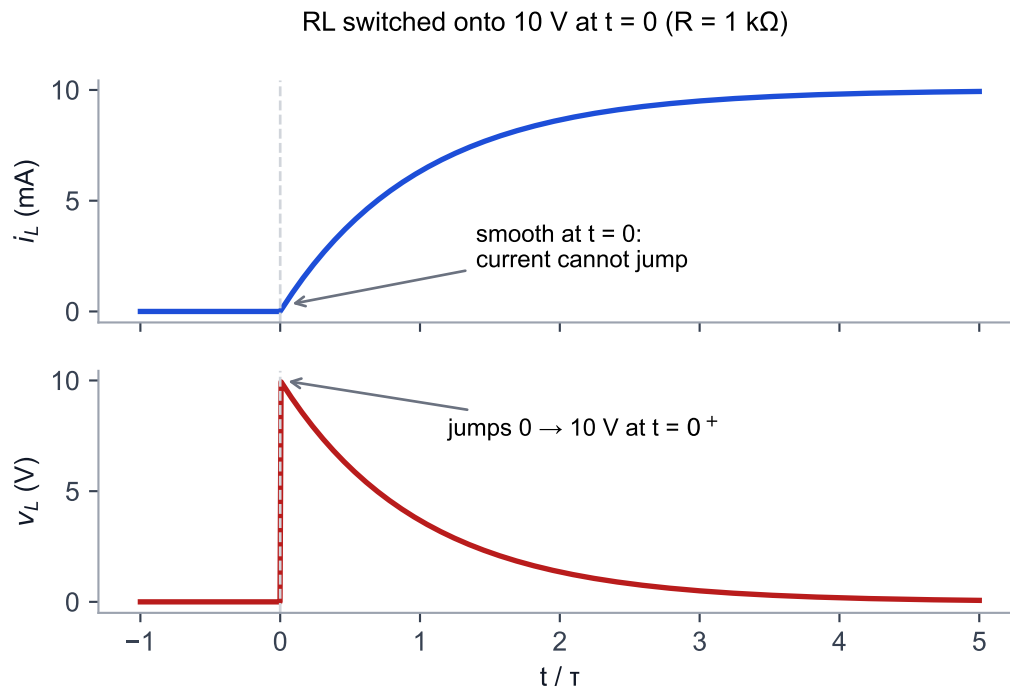
Class p.3 writes the inductor law as " $d/dt (Ci) = C di/dt$ ". It is **L**: $v = d(Li)/dt = \mathbf{L di/dt}$. The side box on the same page ($\psi = Li$) confirms it.

5 • What can't jump

PREDICT

A switch connects 10 V to an R–L circuit at $t = 0$. Just after the switch closes, which one can jump instantly, the inductor's current or its voltage?

- **Inductor current cannot change instantly.** A jump would need $di/dt = \infty$, so $v = L \cdot di/dt = \infty$.
- **Capacitor voltage cannot change instantly.** A jump would need $i = C \cdot dv/dt = \infty$.
- A **resistor** has no memory: its current and voltage can jump (class p.20).



The inductor current rises smoothly from 0. The inductor voltage jumps from 0 to 10 V at the instant of switching.

CHECK 3

(a) Which quantity of a capacitor may jump at $t = 0$: its voltage or its current? (b) Why does a large spark appear when you open a switch in series with a motor coil?

6 · Lumped vs distributed parameters

PREDICT

A 50 Hz power line is 300 km long. A phone's 2.4 GHz Wi-Fi trace is 3 cm long. Which one is "short" compared with the wave travelling along it?

The class split (set 2 p.1): **lumped** parameters are "visible"; **distributed** parameters are "not visible", a property spread along the conductor.

- **Lumped:** each R, L, C sits at one point, as a separate component. Wires between them are ideal. KVL and KCL hold exactly. Every circuit in this pack is lumped.
- **Distributed:** resistance, inductance and capacitance are **spread along the length** of a conductor (per metre), so no single component exists to point at. Transmission lines, antennas and long cables work this way.

When a circuit can be treated as lumped depends on its size compared with the wavelength $\lambda = \frac{c}{f}$:

Signal	$\lambda = c/f$	A circuit of that size is
50 Hz mains	$3 \times 10^8 / 50 = \mathbf{6000 \text{ km}}$	lumped, even across a city
1 GHz	$3 \times 10^8 / 10^9 = \mathbf{30 \text{ cm}}$	distributed at board scale

Rule of thumb: lumped when the circuit is much smaller than λ (often stated as below $\lambda/10$). For the Predict: the 300 km line is $\lambda/20$, so it still counts as lumped (just). The 3 cm trace is about a quarter of $\lambda = 12.5 \text{ cm}$, so it is distributed.

CHECK 4

Why does KVL around a loop become only approximate when the loop is comparable in size to λ ?

Exam form

- **Active element:** can deliver net energy to the circuit (sources; amplifiers, which draw from a supply).
Passive: cannot (R, L, C).
- **Dependent source:** its value depends on another circuit variable. Four types: VCVS, VCCS, CCVS, CCCS (table in step 3, with symbols).
- **Laws:** $v = iR$, $v = L \, di/dt$, $i = C \, dv/dt$.
- **Continuity:** $i_L(0^+) = i_L(0^-)$ and $v_C(0^+) = v_C(0^-)$.
- **DC steady state:** $L \rightarrow \text{short}$, $C \rightarrow \text{open}$.
- **Lumped:** parameters concentrated in separate elements, circuit $\ll \lambda$. **Distributed:** R, L, C spread along the conductor (transmission lines), circuit comparable to λ .
- Devices: VCVS = op-amp, CCCS = BJT, VCCS = MOSFET, CCVS = transimpedance amplifier.

Traps

- "An inductor stores current" and "a capacitor stores voltage" (class-note wording) $\rightarrow$ the exam answer is **energy in a magnetic / electric field**.
- Writing C for L in $v = L \, di/dt$.
- A dependent source's value is **not** a fixed number: it changes whenever its controlling variable changes.

Self-test

1. Name the four dependent sources and give the unit of each constant.
2. Derive $v = L \, di/dt$ starting from $\psi = N\phi$.
3. Why is an inductor a short circuit at DC steady state?
4. Energy stored in 1 mH carrying 10 mA?
5. True or false: a resistor's current cannot change suddenly.
6. A 10 cm circuit runs at 3 GHz. Lumped or distributed? Show λ .

Answers

Check 1. (a) Passive: it is absorbing energy at that moment. (b) It delivers more signal power to the load than the input provides; the extra comes from its supply.

Check 2. (a) A diamond with an arrow inside, labelled $3I_1$. (b) VCVS; μ is the open-loop (voltage) gain.

Check 3. (a) Its current ($i = C dv/dt$ can jump; v cannot). (b) The coil current cannot stop instantly, so $v = L di/dt$ becomes huge and arcs across the opening contacts.

Check 4. KVL assumes the voltage is the same all along a wire at one instant. When the loop is comparable to λ , the wave takes a real fraction of a period to travel around it, so different points on "the same wire" are at different phases.

Self-test.

1. VCVS (μ , no unit) · CCVS (r , Ω) · VCCS (g , S) · CCCS (β , no unit).
2. $\psi = N\phi \propto i \Rightarrow \psi = Li$; $v = d\psi/dt = d(Li)/dt = L di/dt$.
3. At steady state $di/dt = 0$, so $v = L \cdot 0 = 0$ whatever the current: that is a short.
4. $\frac{1}{2} \times 1 \times 10^{-3} \times (10 \times 10^{-3})^2 = \mathbf{50 \text{ nJ}}$.
5. False: a resistor has no memory.
6. $\lambda = 3 \times 10^8 / 3 \times 10^9 = \mathbf{10 \text{ cm}}$: the circuit is a whole wavelength long, so **distributed**.

02 — Ohm's law, dividers, power and energy

Class notes p.1, p.4–5, set 2 p.4–6 · MTE syllabus item 2 · 4 board questions · needs 01

Map

1 Voltage division

→

2 Current division

→

3 Division with AC

→

4 Power and energy

→

5 Ladder reduction

The questions this file answers

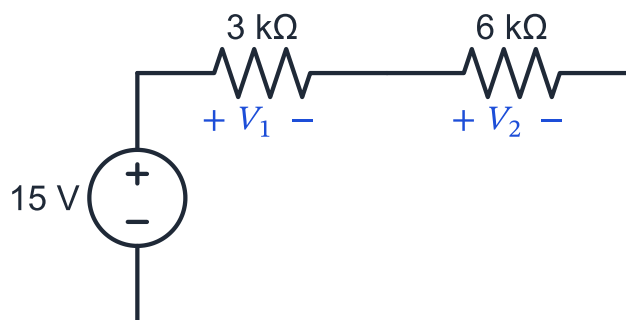
- **Class p.4** 15 V across 3 kΩ and 6 kΩ in series. Find V_1 and V_2 .
- **Class p.5** A 12 A source feeds 3 kΩ || 6 kΩ || 4 kΩ. Find the current in each.
- **Class p.5** $v = 10 \sin(2000t + 30^\circ)$ V across 3 kΩ and 7 kΩ in series. Find v_1 and v_2 .
- **Class p.1** An infinite ladder: 3 kΩ shunt, 5 kΩ series, 7 kΩ shunt, 11 kΩ series, 13 kΩ shunt, ... Find R_{eq} .

Build

1 · Voltage division

CLASS P.4

15 V is applied across 3 kΩ and 6 kΩ in series. Find the voltage across each.



Series resistors carry the same current, so each one's voltage is proportional to its resistance.

GUESS FIRST

Which resistor gets more voltage, and roughly how much?

Series means **one current**: $I = \frac{V}{R_1 + R_2}$. Each voltage is that current times its own resistance:

$$V_1 = V \cdot \frac{R_1}{R_1 + R_2} \quad V_2 = V \cdot \frac{R_2}{R_1 + R_2}$$

$V_1 = 15 \times \frac{3}{9} = 5 \text{ V}$, $V_2 = 15 \times \frac{6}{9} = 10 \text{ V}$. They add back to 15 V (KVL).

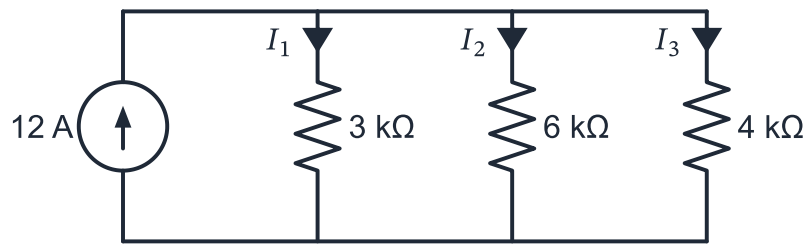
CHECK 1

12 V across 1 k Ω , 2 k Ω , 3 k Ω in series. Voltage across the 2 k Ω ?

2 • Current division

CLASS P.5

A 12 A current source feeds 3 k Ω , 6 k Ω and 4 k Ω in parallel. Find I_1 , I_2 , I_3 .



Parallel branches share one voltage, so the current splits in inverse proportion to resistance.

GUESS FIRST

Which branch carries the most current? Rank the three before calculating.

Parallel means **one voltage**. For two branches that gives the class rule, where **each branch gets the other resistor on top**:

$$I_1 = I \cdot \frac{R_2}{R_1 + R_2} \quad I_2 = I \cdot \frac{R_1}{R_1 + R_2}$$

For three branches, the professor's method is to merge the other two branches into one, then apply the two-branch rule:

Branch	The other two, merged	Current
3 k Ω	$6 \parallel 4 = 2.4 \text{ k}\Omega$	$12 \times \frac{2.4}{2.4 + 3} = 5.33 \text{ A}$
6 k Ω	$3 \parallel 4 = \frac{12}{7} \text{ k}\Omega$	$12 \times \frac{12/7}{12/7 + 6} = 2.67 \text{ A}$
4 k Ω	$3 \parallel 6 = 2 \text{ k}\Omega$	$12 \times \frac{2}{2 + 4} = 4 \text{ A}$

Check: $5.33 + 2.67 + 4 = 12 \text{ A}$ (KCL). Faster in one line: the current in each branch is proportional to its **conductance** $1/R$, i.e. to $1/3 : 1/6 : 1/4 = 4 : 2 : 3$, so 12 A splits as $12 \times \frac{4}{9}$, $12 \times \frac{2}{9}$, $12 \times \frac{3}{9}$.

LABEL IN THE CLASS NOTES

The side note " $I_2 + I_3 = 6 \times 4 / 10 = 2.4$ " is the **resistance** $6 \parallel 4 = 2.4 \text{ k}\Omega$, not a current.

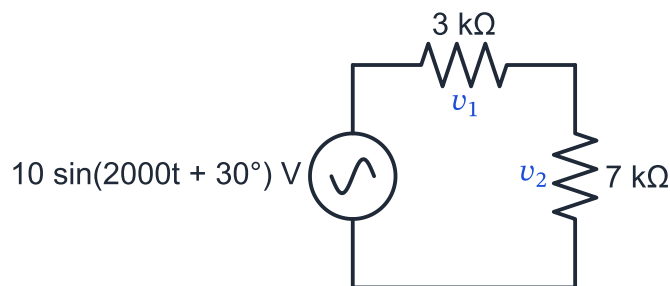
CHECK 2

(a) 10 mA into $1 \text{ k}\Omega \parallel 4 \text{ k}\Omega$: current in the $1 \text{ k}\Omega$? (b) Why does the *smaller* resistor take the *larger* current?

3 • Division works for AC too

CLASS P.5

$v = 10 \sin(2000t + 30^\circ) \text{ V}$ is applied across $3 \text{ k}\Omega$ and $7 \text{ k}\Omega$ in series. Find v_1 and v_2 .

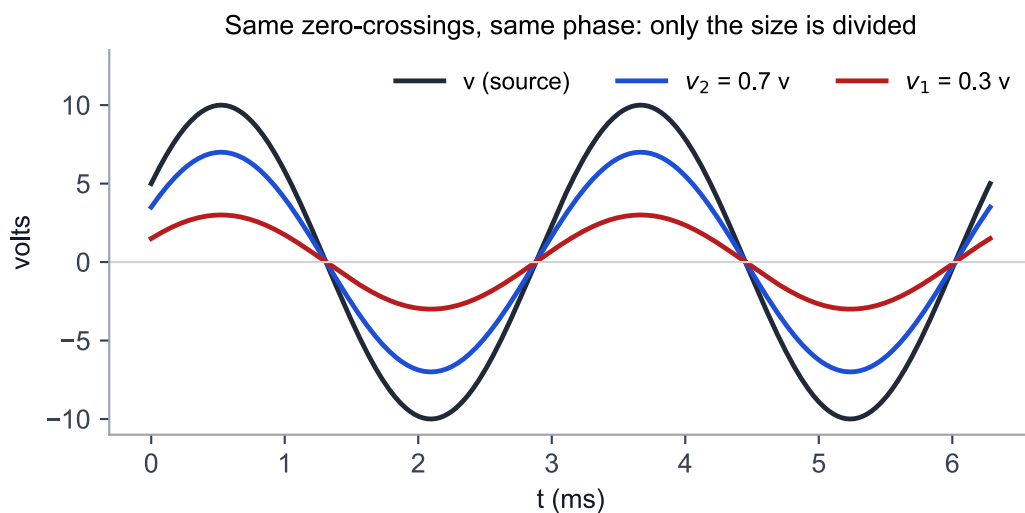


GUESS FIRST

Does v_1 have the same phase angle as the source, or a different one?

A resistor's $v = iR$ holds **at every instant**, so the divider ratio applies to the whole waveform. Resistors do not shift phase:

$$v_1 = 0.3 \times 10 \sin(2000t + 30^\circ) = 3 \sin(2000t + 30^\circ) \text{ V} \quad v_2 = 7 \sin(2000t + 30^\circ) \text{ V}$$



All three waves cross zero together: same frequency, same phase. Only the amplitude is divided.

This holds only because both elements are **resistors**. With an L or C in the divider, the ratio becomes complex and shifts the phase. That case uses impedances (05 §6, 06 §6).

4 · Power and energy: the sign convention

PREDICT

A 12 V battery drives 1 mA. Another circuit forces 1 mA *into* the battery's + terminal (charging it). In which case does the battery absorb power?

Passive sign convention. Label each element's voltage + / -. If the current arrow **enters the + terminal**, then

$$p = v \cdot i \quad \text{is the power ABSORBED by the element}$$

- $p > 0$: the element **absorbs** power (a resistor, always; a battery being charged).
- $p < 0$: the element **delivers** power (a source driving a circuit).
- If the current **leaves** the + terminal, the absorbed power is $p = -v \cdot i$.

Units: $V \times mA = mW$, so $12 V \times 1 mA = 12 mW$. For the Predict: current entering + $\Rightarrow$ the battery absorbs 12 mW (charging); current leaving + $\Rightarrow$ it delivers 12 mW.

Energy is power over time:

$$w = \int p \, dt \quad (\text{joules; constant power: } w = P \cdot t)$$

Element	Power absorbed	Energy
R	$p = i^2 R = \frac{v^2}{R} \geq 0$	all turned into heat
L	$p = Li \cdot \frac{di}{dt}$	stored: $\frac{1}{2} Li^2$
C	$p = Cv \cdot \frac{dv}{dt}$	stored: $\frac{1}{2} Cv^2$

Conservation (Tellegen): in any circuit the absorbed powers add to zero. What the sources deliver, the other elements absorb. 03 uses this as the check on every loop and node answer.

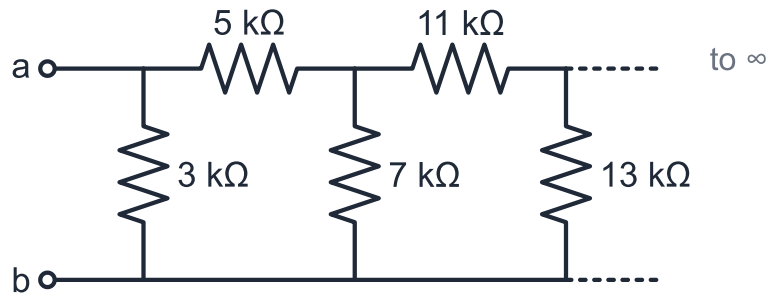
CHECK 3

A 5 V source has 2 mA **leaving** its + terminal. Absorbed power? Is it delivering or absorbing?

5 · Series–parallel reduction: the infinite ladder

CLASS P.1 · THE FIRST QUESTION OF THE COURSE

Find R_{eq} of the infinite ladder: 3 k Ω shunt, 5 k Ω series, 7 k Ω shunt, 11 k Ω series, 13 k Ω shunt, ... ∞ .



Read from the far end: each shunt sits in parallel with everything to its right; each series resistor adds.

GUESS FIRST

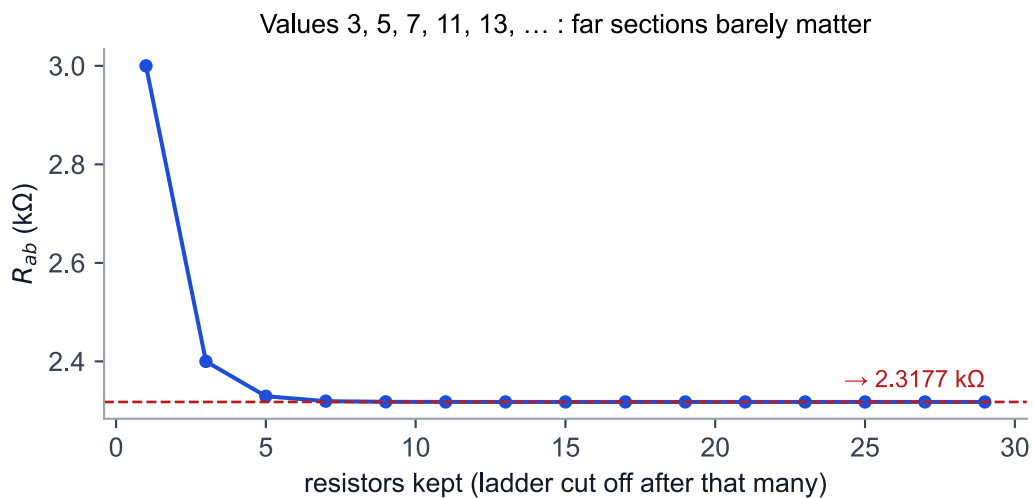
The first resistor is 3 kΩ, directly across a–b. Can R_{eq} be more than 3 kΩ?

Rule: reduce from the far end towards the terminals. Written out, the ladder is:

$$R_{eq} = 3 \parallel (5 + 7 \parallel (11 + 13 \parallel (...))) \quad \text{k}\Omega$$

The Guess answer: the 3 kΩ is in parallel with everything else, so $R_{eq} < 3 \text{ k}\Omega$, always.

The values 3, 5, 7, 11, 13 are the **prime numbers**. Assuming that pattern continues (the notes stop at 13), cut the ladder off after more and more resistors:



3 resistors give 2.400 kΩ, 5 give 2.329, 9 give 2.318. The far sections are huge resistors hidden behind parallel paths, so they barely change the answer.

$R_{eq} \approx 2.318 \text{ k}\Omega$ (converged; confirmed by computer). The professor may intend a different continuation, so ask.

THE STANDARD TRICK, AND WHY IT FAILS HERE

For a ladder whose sections are **all identical** (series R, shunt R, repeated forever), cutting off the first section leaves the **same** infinite ladder. So $R_{eq} = R + (R \parallel R_{eq})$. Solving the quadratic gives $R_{eq} = R(1 + \sqrt{5})/2 \approx 1.618 R$.

Here every section is different, so the rest of the ladder never looks like the whole, and you must truncate.

Exam form

- **VDR:** $V_k = V \cdot \frac{R_k}{R_{total}}$. **CDR (two branches):** $I_1 = I \cdot \frac{R_2}{R_1 + R_2}$.
- Division holds for any waveform when every element is a resistor. Phase is unchanged.
- **Power:** $p = vi$ absorbed when i enters $+$. $p > 0$ absorbs, $p < 0$ delivers. Energy $w = \int p \, dt$; $\frac{1}{2}Li^2$, $\frac{1}{2}Cv^2$.
- **Ladder:** reduce from the far end. Identical infinite ladder: $R_{eq} = R + R \parallel R_{eq}$.

Traps

- CDR with the **same** resistor on top ($I \cdot R_1 / (R_1 + R_2)$ for I_1) is the most common slip: the answer comes out swapped.
- Power sign: a source whose current **leaves** $+$ delivers power, so its absorbed power is **negative**.
- Forgetting that R_{eq} of a network with a shunt resistor across the terminals must be smaller than that resistor.

Self-test

1. 20 V across 2 k Ω and 8 k Ω in series. Both voltages?
2. 9 mA into 3 k $\Omega \parallel$ 6 k Ω . Both currents?
3. $v = 6 \sin(100t)$ V across 1 k Ω and 2 k Ω in series. v across the 2 k Ω ?
4. 3 mA flows through a 2 k Ω resistor for 10 s. Power, and energy turned into heat?
5. An infinite ladder of identical 1 k Ω sections (series then shunt). R_{eq} ?

Answers

Check 1. $12 \times 2/6 = 4 \text{ V}$.

Check 2. (a) $10 \times 4/5 = 8 \text{ mA}$. (b) Same voltage across both, and $I = V/R$, so the smaller R takes more current.

Check 3. $p = -5 \times 2 = -10 \text{ mW}$ absorbed: it **delivers** 10 mW.

Self-test.

1. **4 V** and **16 V**.

2. $3 \text{ k}\Omega$: $9 \times 6/9 = 6 \text{ mA}$; $6 \text{ k}\Omega$: **3 mA**.

3. $\frac{2}{3} \times 6 \sin(100t) = 4 \sin(100t) \text{ V}$.

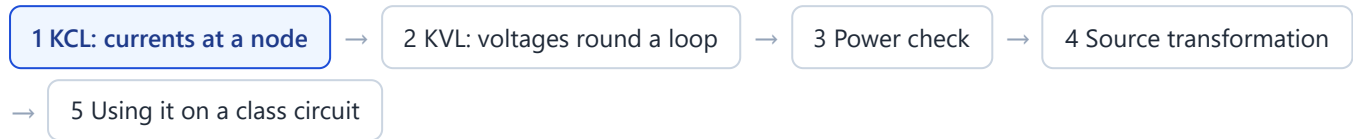
4. $P = (3 \text{ mA})^2 \times 2 \text{ k}\Omega = 18 \text{ mW}$; $w = 18 \text{ mW} \times 10 \text{ s} = 0.18 \text{ J}$.

5. $R(1 + \sqrt{5})/2 = 1.618 \text{ k}\Omega$.

03 — KVL, KCL and source transformation

Class notes p.4–5, set 2 p.4, p.11 · MTE syllabus item 3 · 1 board question + 1 textbook problem · needs 02

Map



The questions this file answers

- **textbook** A loop with 12 V, 2 k Ω , an opposing 4 V source and 4 k Ω : find i , each voltage, and each element's power.
- **Class p.4** Source transformation: V in series with R $\Leftrightarrow$ V/R in parallel with R.
- **Class p.7 · set 2 p.8** The superposition circuit again, solved in three lines by source transformation.

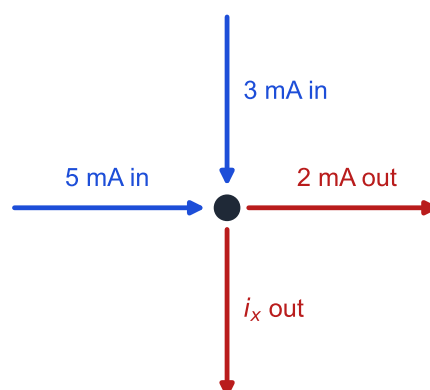
The professor used both laws inside other methods (KCL in the nodal working on set 2 p.11; KVL in every divider). This file states them on their own, as the syllabus asks.

Build

1 · KCL: what goes into a node comes out

PREDICT

5 mA and 3 mA flow into a node. 2 mA flows out along a third wire. What flows in the fourth wire, and which way?



One node, four wires.

Kirchhoff's current law (KCL): the algebraic sum of the currents at any node is zero. Equivalently, **total current in = total current out**.

$$\sum i_{\text{in}} = \sum i_{\text{out}}$$

Why it holds: charge cannot pile up at a node (a node stores no charge). So every coulomb that arrives per second must leave.

For the Predict: $5 + 3 = 2 + i_x$, so **$i_x = 6 \text{ mA}$, out**. If you had drawn i_x pointing in, you would get $i_x = -6 \text{ mA}$: the same answer. A minus sign only means the real current flows against your arrow.

You already used KCL in 02: the three branch currents of the class p.5 question add back to the source, $5.33 + 2.67 + 4 = 12 \text{ A}$.

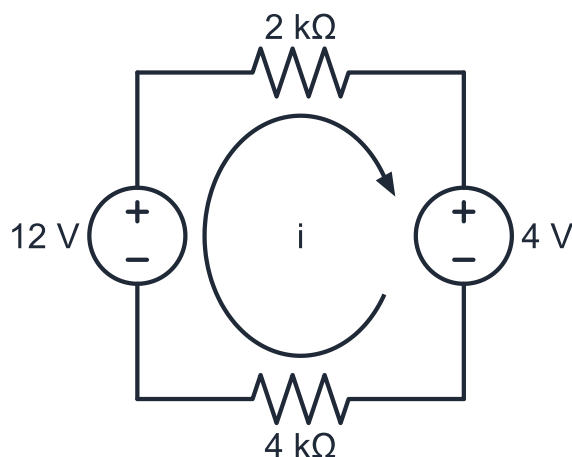
CHECK 1

Currents into a node: 4 mA , -1 mA , i . Out of it: 7 mA . Find i .

2 · KVL: round a loop, the voltages add to zero

TEXTBOOK

Find the loop current i , the voltage across each resistor, and the power of every element.



The 4 V source has $+$ at the top, so it pushes against the 12 V source.

GUESS FIRST

Which source "wins", and is the current in the direction of the arrow shown?

Kirchhoff's voltage law (KVL): the algebraic sum of the voltages around any closed loop is zero.

$$\sum v_{\text{rises}} = \sum v_{\text{drops}} \quad (\text{round any closed loop})$$

Why it holds: voltage is energy per unit charge. A charge that travels round a loop and returns to its start has the same energy it began with.

Method. Go round the loop in the direction of i (clockwise). A source traversed from $-$ to $+$ is a **rise**. A resistor traversed along i is a **drop** of iR . A source traversed from $+$ to $-$ is a **drop**.

- 12 V source, bottom to top (– to +): rise of 12 V.
- 2 kΩ: drop $2i$. 4 V source, top to bottom (+ to –): drop 4 V. 4 kΩ: drop $4i$.

$$12 = 2i + 4 + 4i \quad \Rightarrow \quad i = \frac{8}{6} = \frac{4}{3} \text{ mA} \approx 1.33 \text{ mA}$$

Voltages: $v_{2k\Omega} = 2 \times \frac{4}{3} = \mathbf{2.67 \text{ V}}$, $v_{4k\Omega} = \mathbf{5.33 \text{ V}}$. Check round the loop: $12 = 2.67 + 4 + 5.33 \checkmark$.

The Guess answer: the 12 V source wins, and $i > 0$ means the current does flow clockwise.

CHECK 2

Swap the 4 V source (+ at the bottom, so it now helps). Find i .

3 • The power check (02 §4, in use)

Sign rule from 02: current **entering** + means the element absorbs $p = vi$.

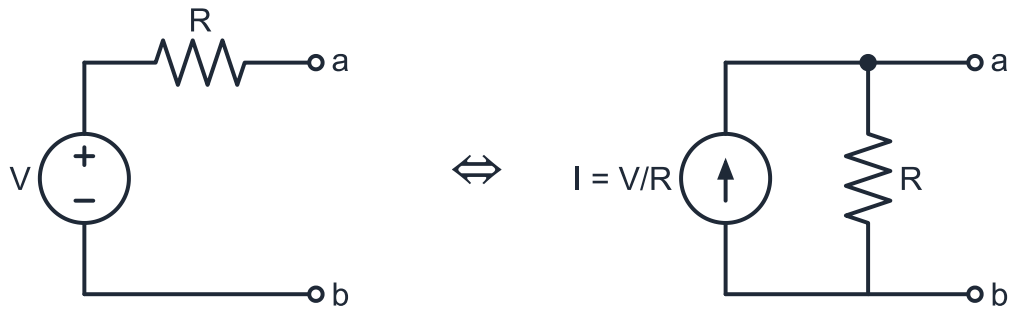
Element	Current enters its...	Absorbed power
12 V source	– terminal (current leaves +)	$-12 \times \frac{4}{3} = \mathbf{-16 \text{ mW}}$ (delivers 16 mW)
4 V source	+ terminal	$+4 \times \frac{4}{3} = \mathbf{+5.33 \text{ mW}}$ (it is being charged)
2 kΩ	+ (always, for a resistor)	$(\frac{4}{3})^2 \times 2 = \mathbf{+3.56 \text{ mW}}$
4 kΩ	+	$(\frac{4}{3})^2 \times 4 = \mathbf{+7.11 \text{ mW}}$
Sum		$-16 + 5.33 + 3.56 + 7.11 = \mathbf{0 \checkmark}$

Delivered = absorbed. Use this as the last line of any loop or node answer: if the powers do not add to zero, something upstream is wrong.

4 • Source transformation

PREDICT

A 10 V source with 2 kΩ in series. Short its terminals: what current flows? Leave them open: what voltage appears? Could a current source with a resistor in parallel give the same two answers?



Both give the same open-circuit voltage (V) and the same short-circuit current (V/R), so no outside circuit can tell them apart.

A **voltage source V in series with R** is equivalent to a **current source $I = V/R$ in parallel with the same R** (class p.4; set 2 p.4 writes $I = V/R$ under the current source), as seen from the terminals a–b.

For the Predict: short-circuit current = $10/2k = \mathbf{5\text{ mA}}$; open-circuit voltage = $\mathbf{10\text{ V}}$. So it is equivalent to $5\text{ mA} \parallel 2\text{ k}\Omega$, and $5\text{ mA} \times 2\text{ k}\Omega = 10\text{ V}$, the same terminal behaviour.

It is the bridge between Thévenin and Norton in 06: $I_N = V_{Th} / R_{Th}$.

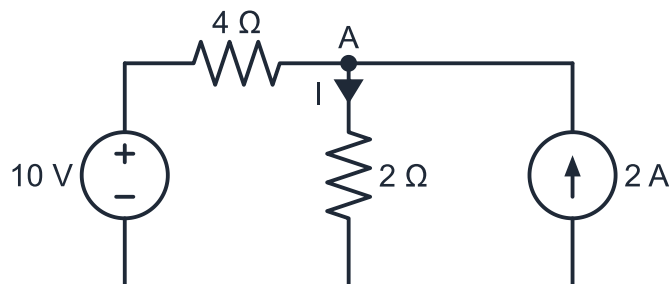
ONLY WORKS WITH A RESISTOR

An **ideal** voltage source (no series R) has no current-source equivalent: $V/0$ is undefined. Same for an ideal current source with no parallel R .

5 · Using it on a class circuit

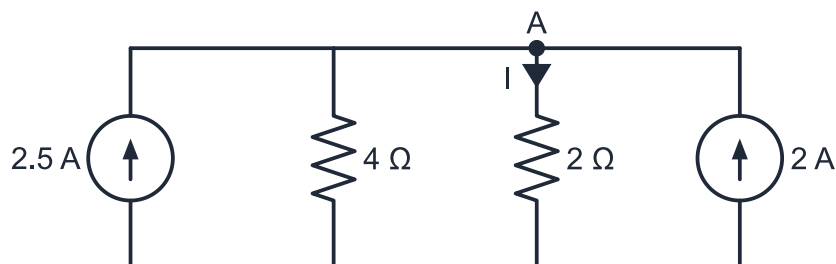
CLASS P.7 · SET 2 P.8

Find I in the superposition circuit (10 V , $4\text{ }\Omega$, $2\text{ }\Omega$, 2 A) without superposition.



The class circuit (05 §1).

Transform 10 V in series with $4\text{ }\Omega$ into $\frac{10}{4} = \mathbf{2.5\text{ A}}$ in parallel with $4\text{ }\Omega$:



Now everything is in parallel between node A and the bottom rail.

Both current sources push up into A: $2.5 + 2 = 4.5$ A. It flows into $4\ \Omega \parallel 2\ \Omega = \frac{4}{3}\ \Omega$, so $V_A = 4.5 \times \frac{4}{3} = \mathbf{6\ V}$ and $\mathbf{I = \frac{6}{2} = 3\ A}$, the professor's answer (05 §1) in three lines.

CHECK 3

Convert 3 mA in parallel with 4 k Ω into a voltage-source form.

Exam form

- **KCL:** Σ currents into a node = Σ currents out (charge is conserved; a node stores no charge).
- **KVL:** Σ voltage rises = Σ voltage drops round any closed loop (energy is conserved).
- **Sign rule:** a negative answer means the real direction is opposite to the arrow you drew. Keep your arrow; report the sign.
- **Power balance:** Σ absorbed power = 0 over the whole circuit.
- **Source transformation:** V in series with R $\Leftrightarrow$ V/R in parallel with R. Not for ideal sources.

Traps

- Changing sign conventions half-way round a loop. Fix the direction once and apply the rise/drop rule to every element.
- Source transformation keeps R **the same size**. It changes from series to parallel, not in value.
- Transforming a source whose resistor is the one you are asked about: that resistor disappears into the equivalent. Keep the element you need outside the transformation.

Self-test

1. State KCL and KVL, each with the conservation law behind it.
2. A node has 10 mA in and three outputs: 2 mA, 5 mA, i. Find i.
3. A loop has 9 V, 1 k Ω and 2 k Ω . Find i and check the power balance.
4. Transform 12 V in series with 3 k Ω .
5. Why is a negative answer from KCL not a mistake?

Answers

Check 1. $4 + (-1) + i = 7 \Rightarrow i = 4 \text{ mA}$ into the node.

Check 2. $12 + 4 = 6i \Rightarrow i = 8/3 \approx 2.67 \text{ mA}$.

Check 3. $3 \text{ mA} \times 4 \text{ k}\Omega = 12 \text{ V}$ in series with $4 \text{ k}\Omega$.

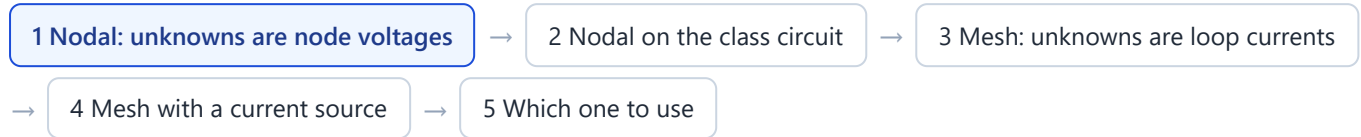
Self-test.

1. KCL: $\sum i_{\text{in}} = \sum i_{\text{out}}$ at a node (conservation of charge). KVL: $\sum v$ round a closed loop = 0 (conservation of energy).
2. $10 = 2 + 5 + i \Rightarrow i = 3 \text{ mA}$.
3. $i = 9/3\text{k} = 3 \text{ mA}$. Source delivers $9 \times 3 = 27 \text{ mW}$; resistors absorb $9 + 18 = 27 \text{ mW}$ ✓.
4. **4 mA in parallel with 3 k Ω .**
5. It only means the current flows opposite to the arrow you chose; the magnitude is still right.

04 — Nodal and mesh (loop) analysis

Set 2 p.11–12 (nodal, used by the professor) · MTE syllabus item 4 · 1 board circuit + 1 textbook problem · needs 03

Map



The questions this file answers

- **textbook** 10 V, 2 k Ω , 4 k Ω (shared), 2 k Ω , 4 V: find every current by nodal analysis, then by mesh analysis.
- **Class p.7 · set 2 p.8** The superposition circuit by nodal and by mesh (answer known: $I = 3$ A).
- **Set 2 p.11–12** The professor's nodal working for V_{Th} : see 06 §2, which uses this file's method.

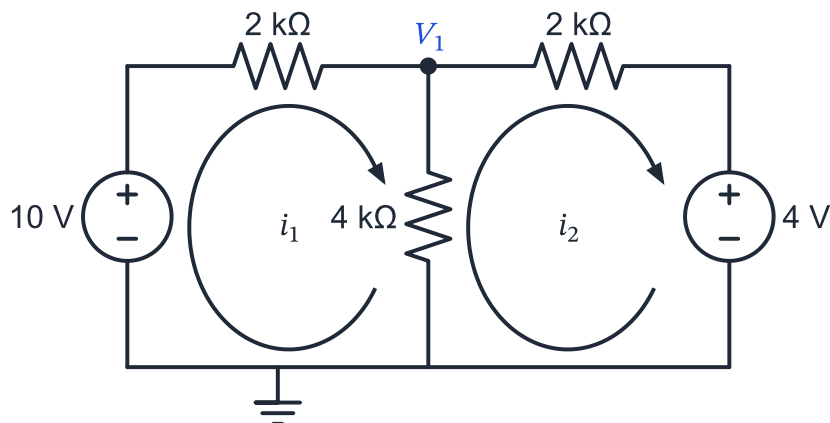
Both methods are KCL (nodal) or KVL (mesh) from 03, written in a fixed order so that nothing gets missed.

Build

1 · Nodal analysis

TEXTBOOK

Find V_1 , and the current in every branch.



Two sources, three resistors. One ground at the bottom; one unknown node, V_1 .

GUESS FIRST

How many unknowns do you need if the unknowns are node voltages? And if they are loop currents?

Procedure. 1. Pick a **reference node** (ground, 0 V). Usually the node with the most connections. 2. Name every other node's voltage: $V_1, V_2, \dots$. A node joined to ground only by a voltage source is known already. 3. At each unknown node write KCL with **every current written as leaving**: $\frac{V_{\text{node}} - V_{\text{other}}}{R}$. 4. Solve the equations.

Here the only unknown is V_1 . Currents leaving it through the left 2 k Ω , the 4 k Ω and the right 2 k Ω (units V, k Ω , mA):

$$\frac{V_1 - 10}{2} + \frac{V_1}{4} + \frac{V_1 - 4}{2} = 0$$

Multiply by 4: $2V_1 - 20 + V_1 + 2V_1 - 8 = 0 \Rightarrow 5V_1 = 28 \Rightarrow \mathbf{V_1 = 5.6 \text{ V}}$.

Branch	Current	Value
left 2 k Ω , to the right	$\frac{10 - 5.6}{2}$	2.2 mA
4 k Ω , downwards	$\frac{5.6}{4}$	1.4 mA
right 2 k Ω , to the right	$\frac{5.6 - 4}{2}$	0.8 mA

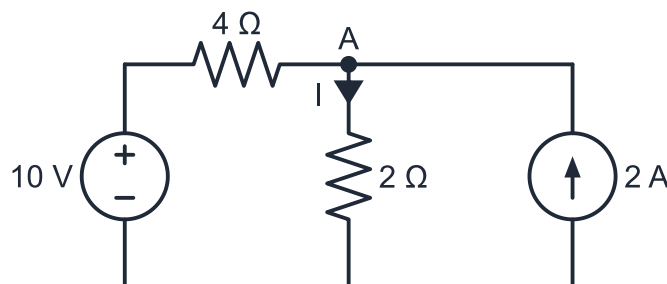
KCL check at V_1 : $2.2 = 1.4 + 0.8 \checkmark$.

The Guess answer: nodal needs (**nodes – 1**) unknowns, here **1**. Mesh needs one per window, here **2**.

2 · Nodal on the class circuit

CLASS P.7 · SET 2 P.8

Find I (down through the 2 Ω) by nodal analysis.



One unknown, V_A . Currents leaving A: through the 4 Ω , $\frac{V_A - 10}{4}$; through the 2 Ω , $\frac{V_A}{2}$; the 2 A source pushes **into** A, so it counts as -2 leaving:

$$\frac{V_A - 10}{4} + \frac{V_A}{2} - 2 = 0 \quad \Rightarrow \quad V_A = 6 \text{ V}, \quad I = \frac{6}{2} = 3 \text{ A}$$

Same 3 A as superposition (05) and source transformation (03). A current source simply puts a known current into the KCL equation.

With a dependent source (06 §2, set 2 p.11): write the source's value in terms of the node voltages first (there, $I = \frac{10 - V_A}{2}$), substitute, and the equation again has one unknown. That is exactly the professor's nodal working.

CHECK 1

In the textbook circuit, change the 4 V source to 0 V (a short). Write the nodal equation and find V_1 .

3 • Mesh analysis

TEXTBOOK

Same circuit. Find the mesh currents i_1 , i_2 .

Procedure. 1. Give each **window** (mesh) of the circuit a clockwise current: i_1 , i_2 , ... 2. A resistor shared by two meshes carries the **difference**: $(i_1 - i_2)$ in mesh 1's direction. 3. Write KVL round each mesh: $\Sigma \text{ rises} = \Sigma \text{ drops}$. 4. Solve.

Mesh 1 (10 V rise; drops across the left 2 k Ω and the shared 4 k Ω):

$$10 = 2i_1 + 4(i_1 - i_2) \quad \Rightarrow \quad 6i_1 - 4i_2 = 10$$

Mesh 2 (clockwise, the current goes **down** through the 4 V source from + to –, a drop):

$$0 = 4(i_2 - i_1) + 2i_2 + 4 \quad \Rightarrow \quad -4i_1 + 6i_2 = -4$$

Solve (determinant $6 \cdot 6 - 4 \cdot 4 = 20$): **$i_1 = 2.2 \text{ mA}$, $i_2 = 0.8 \text{ mA}$** . The 4 k Ω carries $i_1 - i_2 = \mathbf{1.4 \text{ mA}}$ down, and $1.4 \times 4 = \mathbf{5.6 \text{ V}}$: the same answers as nodal $\checkmark$.

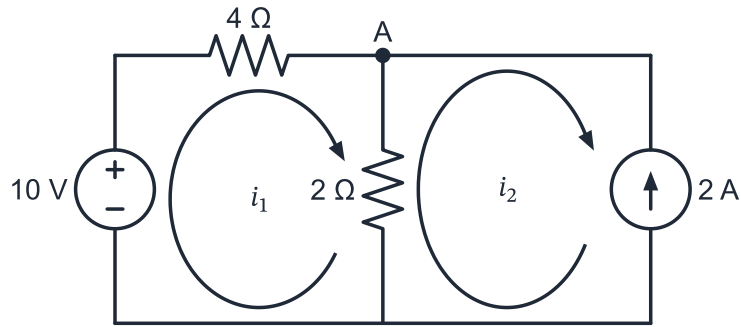
READING A MESH ANSWER

A mesh current is a **bookkeeping** current. The current in a resistor that belongs to one mesh is that mesh current; in a shared resistor it is the difference.

4 • Mesh with a current source

CLASS P.7 • SET 2 P.8

Find I by mesh analysis.



The 2 A source (arrow up) sits on the outer edge of mesh 2 only.

A current source on the **outside edge** of a mesh fixes that mesh current. Mesh 2 goes **down** through the source, but the source pushes 2 A **up**:

$$i_2 = -2 \text{ A}$$

Mesh 1: $10 = 4i_1 + 2(i_1 - i_2) = 6i_1 + 4 \Rightarrow i_1 = 1 \text{ A}$.

The 2Ω carries $i_1 - i_2 = 1 - (-2) = 3 \text{ A downwards}$ ✓. The 4Ω carries $i_1 = 1 \text{ A}$, which agrees with nodal:

$$\frac{10 - 6}{4} = 1 \text{ A}.$$

If a current source is **shared** by two meshes, its current fixes their difference, and you write one KVL round the combined loop that avoids the source (a **supermesh**). The nodal twin: a voltage source between two unknown nodes fixes their difference, and you write one KCL round both (a **supernode**).

CHECK 2

Why can't you write a KVL equation for mesh 2 in this circuit the normal way?

5 · Which one to use

Use nodal when...	Use mesh when...
fewer nodes than meshes	fewer meshes than nodes
the circuit has current sources	the circuit has voltage sources in series with resistors
you need voltages (e.g. V_{Th})	you need loop currents
the circuit is non-planar (wires cross)	the circuit is planar (mesh needs that)

Exam form

Nodal. (1) Choose a reference node. (2) Label the other node voltages. (3) KCL at each unknown node, currents written as leaving: $(V_n - V_m)/R$. (4) Solve. Unknowns = nodes – 1 (less any fixed by voltage sources). Voltage source between two unknown nodes → supernode.

Mesh. (1) One clockwise current per window. (2) Shared element carries the difference. (3) KVL per mesh. (4) Solve. Unknowns = number of windows. Current source on an outer edge fixes that mesh current; a shared one → supermesh.

Dependent sources. Write the controlling variable in terms of the unknowns, substitute, then solve.

Traps

- Mixing "leaving" and "entering" within one nodal equation.
- In mesh analysis, writing the shared resistor's voltage as $R \cdot i_1$ instead of $R(i_1 - i_2)$.
- Mesh analysis on a circuit with a current source: trying to write the source's voltage as "0". It is unknown; use the source to fix the mesh current instead.
- Unit mix: with $k\Omega$, currents come out in **mA** (the set 2 p.12 working mixes $k\Omega$ and A; see 06 §2).

Self-test

1. How many equations does nodal analysis need for a circuit with 5 nodes and no voltage sources?
2. In the textbook circuit, find the power delivered by the 10 V source.
3. Write the two mesh equations if the 4 V source were reversed (+ at the bottom).
4. What is a supernode, and when do you need one?
5. Solve the class circuit by mesh if the 2 A source pointed down.

Answers

Check 1. $\frac{V_1 - 10}{2} + \frac{V_1}{4} + \frac{V_1}{2} = 0 \Rightarrow 5V_1 = 20 \Rightarrow \mathbf{V_1 = 4\text{ V.}}$

Check 2. The voltage across a current source is not known in advance (it is whatever the circuit makes it), so the KVL equation would contain an extra unknown. The source already gives i_2 directly.

Self-test.

1. **4** (nodes - 1).
2. $10\text{ V} \times 2.2\text{ mA} = \mathbf{22\text{ mW}}$ (current leaves its + terminal: it delivers).
3. $6i_1 - 4i_2 = 10$ and $-4i_1 + 6i_2 = \mathbf{+4}$ (now a rise) $\Rightarrow i_1 = 3.8\text{ mA}$, $i_2 = 3.2\text{ mA}$.
4. A voltage source connected between two non-reference nodes: enclose both nodes, write one KCL for the pair plus $V_a - V_b = V_s$.
5. $i_2 = +2\text{ A}$; $10 = 6i_1 - 4 \Rightarrow i_1 = 7/3\text{ A}$; $I = i_1 - i_2 = \mathbf{1/3\text{ A}}$ (matches 05 Check 1b).

05 — Superposition theorem

Class notes p.6–9, set 2 p.7–9 · MTE syllabus item 5 (DC and AC) · 2 board questions + 1 textbook · needs 01 (source internal resistances), 02 (dividers)

Map



The questions this file answers

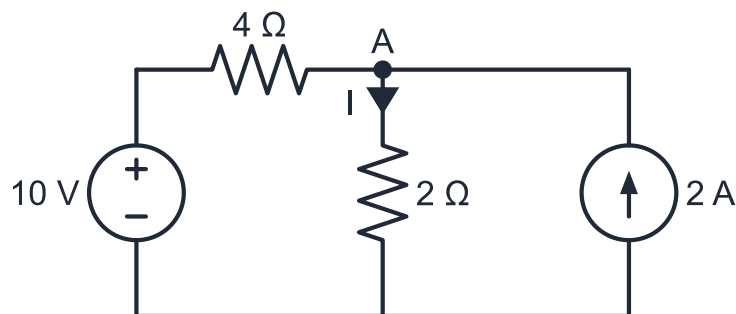
- **Class p.7** Calculate the current I using the superposition theorem (10 V, 2 A, two resistors).
- **Class p.9 · set 2 p.9** Calculate the current I_1 using superposition (two 5 V sources across 1 k Ω).
- **textbook** A DC and an AC source in one circuit: find $i(t)$.

Build

1–3 · The method

CLASS P.7

Calculate the value of current I using the superposition theorem.



The board circuit. I is the current down through the 2 Ω resistor.

UNITS ON THE BOARD

The board labels read "4k" and "2k", but the professor's working ($10 \times 2/6 = 10/3$ V, then $5/3$ A) treats them as **4 Ω and 2 Ω** . The pack uses ohms, so every number stays consistent. A full computer solve of this circuit gives exactly the professor's answer, **3 A**.

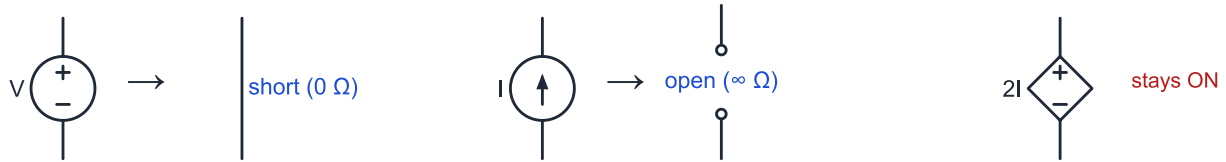
GUESS FIRST

Two sources push current into the 2 Ω . Would you try solving for both at once, or one at a time? If one at a time, what do you do with the other source?

Statement (class p.6): in a **linear** circuit with several independent sources, the response in any element (a current or a voltage) is the **algebraic sum** of the responses produced by **each source acting alone**.

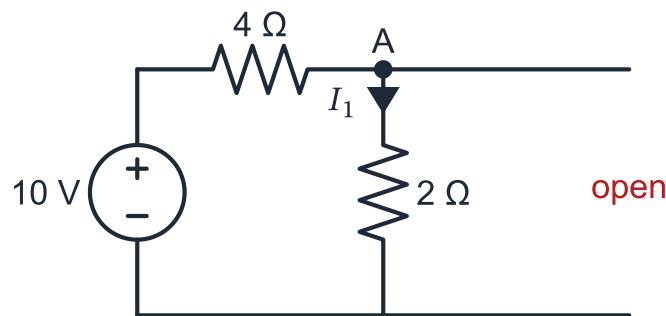
It works because a linear circuit's response is a weighted sum of its source values, so each source's share can be found separately.

Switching a source off = replacing it with its **internal resistance** (class p.7, and 01 step 2):



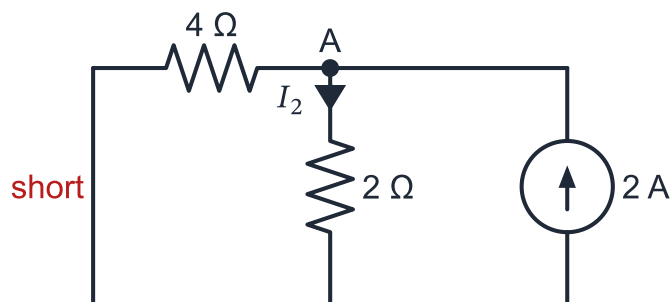
Only independent sources are switched off. A dependent source always stays in the circuit.

Case 1: 10 V alone (2 A source $\rightarrow$ open):



$4\ \Omega$ and $2\ \Omega$ are in series across 10 V. By voltage division, $V_{2\Omega} = 10 \times \frac{2}{6} = \frac{10}{3}$ V, so $I_1 = \frac{10/3}{2} = \frac{5}{3}$ A $\downarrow$.

Case 2: 2 A alone (10 V source $\rightarrow$ short):



With the 10 V shorted, the $4\ \Omega$ runs from A to the bottom rail, so it is now **in parallel** with the $2\ \Omega$. The 2 A splits by current division: $I_2 = 2 \times \frac{4}{4 + 2} = \frac{4}{3}$ A $\downarrow$.

Step 3: add, with signs. Both cases push current **down** through the $2\ \Omega$, so they add:

$$I = I_1 + I_2 = \frac{5}{3} + \frac{4}{3} = 3\text{ A}$$

Cross-check without superposition. One node equation at A: $\frac{10 - V_A}{4} + 2 = \frac{V_A}{2}$, so $V_A = 6 \text{ V}$ and $I = 6/2 = 3 \text{ A}$ ✓.

CHECK 1

(a) In case 2, why is the 4Ω in *parallel* with the 2Ω , when it was in *series* in case 1? (b) If the 2 A source pointed downwards instead, what would I be?

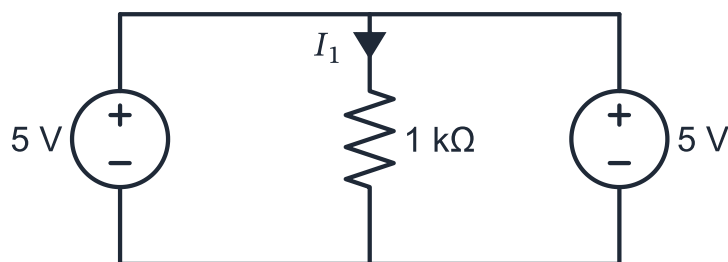
4 • Signs matter

Pick a reference direction for the answer **once**, before the cases. In each case, a current that flows **against** that direction enters the sum as **negative**. The class example happened to have both cases in the same direction.

5 • Where superposition fails

CLASS P.9 • SET 2 P.9

Calculate the value of current I_1 using superposition.



Both sources have + at the top, as drawn in class. Both sit directly across the $1 \text{ k}\Omega$.

GUESS FIRST

Apply superposition exactly by the rule: left source alone (right source $\rightarrow$ its internal resistance), then right source alone. Add. Now look at the circuit: what voltage is actually across the $1 \text{ k}\Omega$?

The professor's working (set 2 p.9), by the rule:

- **Case 1, left 5 V alone.** The right source is replaced by its internal resistance, 0Ω : a **short**. That short sits directly across the $1 \text{ k}\Omega$, so $I_1 = 0$.
- **Case 2, right 5 V alone.** By symmetry the left source becomes a short across the $1 \text{ k}\Omega$: $I_1 = 0$.
- **Sum:** $I_{\text{total}} = 0 + 0 = 0 \text{ A}$, which the professor labels **LIMITATION**.

The true answer. Both ideal sources hold the top rail at 5 V , so the $1 \text{ k}\Omega$ has 5 V across it (KVL round either loop): $I_1 = \frac{5}{1\text{k}} = 5 \text{ mA}$. Superposition gave 0 A . It is wrong here, not slightly off.

Why it fails. In each case the short that replaces the idle source is also placed **across the active source**. An ideal 5 V source shorted by a wire would need infinite current, so each case sub-circuit is impossible, and the "0" it produces means nothing. This is the professor's limitation "sometimes gives false results in ideal circuits"

(set 2 p.8): **ideal voltage sources in parallel** (or ideal current sources in series) cannot be switched off one at a time.

TWO VERSIONS IN THE NOTES

The first set of notes (class p.9) sketched case 1 with the idle source **removed** (an open) instead of shorted. That gives $5 + 5 = 10$ mA, also wrong, and for a second reason: an idle voltage source must be shorted, never opened. The set 2 version applies the rule correctly and shows the real point: the rule itself breaks down here. Learn the set 2 version.

The limitations, complete:

Limitation	Why
Linear circuits only	the sum rule <i>is</i> linearity; a diode breaks it (set 2 p.8: forward / reverse bias, superposition fails; class p.7: non-linear, unilateral)
Not for power	$P = I^2R$ is not linear in I . See below
Dependent sources are never switched off	their value is set by a circuit variable, not by an outside excitation
Ideal sources that conflict	parallel ideal V sources / series ideal I sources: each case gives 0, the truth is 5 mA (Q above)

Why power can't be superposed, using the Q from step 1:

	Current in $2\ \Omega$	Power in $2\ \Omega$
10 V alone	$5/3$ A	$(5/3)^2 \times 2 = 5.56$ W
2 A alone	$4/3$ A	$(4/3)^2 \times 2 = 3.56$ W
"sum of powers"		9.11 W X
actual ($I = 3$ A)	3 A	$3^2 \times 2 = \mathbf{18\ W}$ ✓

Superpose the **currents**, then square.

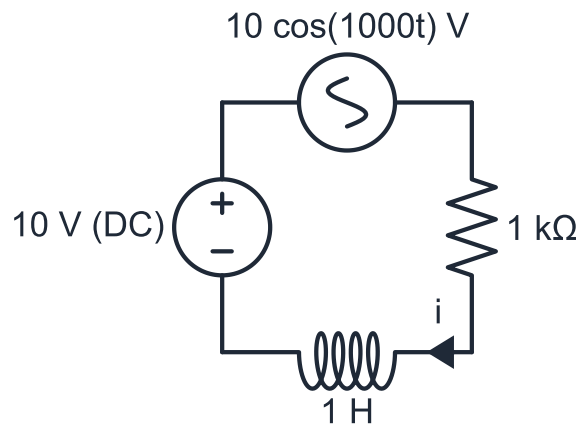
CHECK 2

Why does squaring break superposition? Give the one-line algebra reason.

6 · Superposition with AC sources

TEXTBOOK

A 10 V DC source and a $10 \cos(1000t)$ V source drive $1\ \text{k}\Omega$ and $1\ \text{H}$ in series. Find the steady-state current $i(t)$.



GUESS FIRST

Can you turn both sources into phasors and add them?

No: a phasor only exists **at one frequency**. DC is $\omega = 0$, the other source is $\omega = 1000 \text{ rad/s}$. The impedance of the inductor is different at each ($Z_L = j\omega L$), so the circuit must be solved **once per frequency**, and the answers added **in the time domain**. This is the one case where superposition is not a choice but the only way.

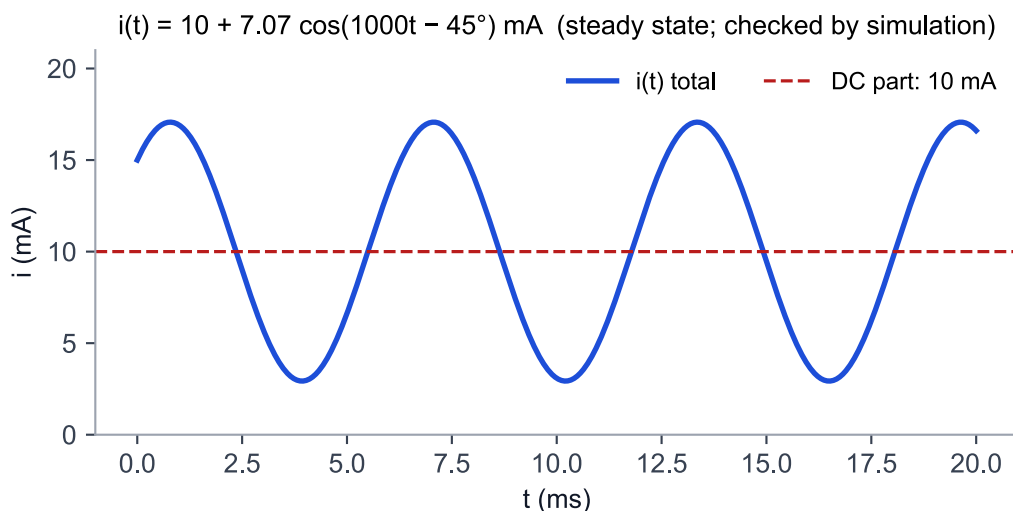
Case 1: DC alone (AC source $\rightarrow$ short). At DC the inductor is a short (0Ω): $i_1 = \frac{10}{1\text{k}} = \mathbf{10 \text{ mA}}$.

Case 2: AC alone (DC source $\rightarrow$ short). At $\omega = 1000$: $Z = R + j\omega L = 1000 + j1000 \Omega = 1414\angle 45^\circ \Omega$.

$$I_2 = \frac{10\angle 0^\circ}{1414\angle 45^\circ} = 7.07\angle -45^\circ \text{ mA} \quad \Rightarrow \quad i_2(t) = 7.07 \cos(1000t - 45^\circ) \text{ mA}$$

Add in time:

$$i(t) = 10 + 7.07 \cos(1000t - 45^\circ) \text{ mA}$$



The AC part rides on the 10 mA DC level. A time simulation of the circuit gives the same curve.

Same frequency? If all sources share one ω , you may add their **phasors** directly, and the result is the same as adding in time. The rule for switching sources off does not change: $V \rightarrow \text{short}$, $I \rightarrow \text{open}$, dependent sources stay.

CHECK 3

Would I_2 change if the inductor were 2 H? Give the new phasor.

Exam form

Statement. In a linear, bilateral network containing more than one independent source, the response in any element equals the algebraic sum of the responses caused by each independent source acting alone, with all other independent sources replaced by their internal resistances (V source $\rightarrow$ short, I source $\rightarrow$ open). Dependent sources remain active.

Procedure. 1. Choose the reference direction of the unknown. 2. Keep one independent source; switch off the others ($V \rightarrow$ short, $I \rightarrow$ open). Keep dependent sources. 3. Find that source's contribution with dividers or series-parallel reduction. 4. Repeat for every independent source. 5. Add the contributions algebraically (with signs).

Limitations. Not for non-linear or unilateral elements · not for power · dependent sources stay on · fails for conflicting ideal sources.

AC. Sources at different frequencies: solve each at its own ω ($Z_L = j\omega L$, $Z_C = 1/j\omega C$), convert each to a time function, add in time. Same frequency: phasors may be added.

Traps

- Removing a voltage source (leaving an open) instead of **shorting** it. The class p.9 sketch does this; set 2 p.9 corrects it.
- Adding phasors of **different** frequencies.
- Switching off a dependent source.
- Adding powers.
- Losing the sign when a contribution flows opposite to the chosen direction.

Self-test

1. State superposition in one sentence, including what happens to the other sources.
2. What is the internal resistance of an ideal current source, and what does it become when switched off?
3. In the step-1 circuit, find the voltage at node A from each source separately, then add.
4. Can superposition find the power in a resistor directly? What do you do instead?
5. A circuit has one independent source and one dependent source. How many superposition cases are there?
6. For the two-5 V circuit, what does superposition give, and what is the true I_1 ?

Answers

Check 1. (a) Shorting the 10 V source connects the 4 Ω 's left end to the bottom rail, so both resistors now join node A to the bottom rail: parallel. (b) $I_2 = -4/3$ A, so $I = 5/3 - 4/3 = \mathbf{1/3 \text{ A}}$.

Check 2. $(I_1 + I_2)^2 = I_1^2 + I_2^2 + 2I_1I_2$. Adding the separate powers loses the $2I_1I_2$ cross term.

Check 3. $Z = 1000 + j2000 = 2236\angle 63.4^\circ \Omega \Rightarrow I_2 = \mathbf{4.47\angle -63.4^\circ \text{ mA}}$.

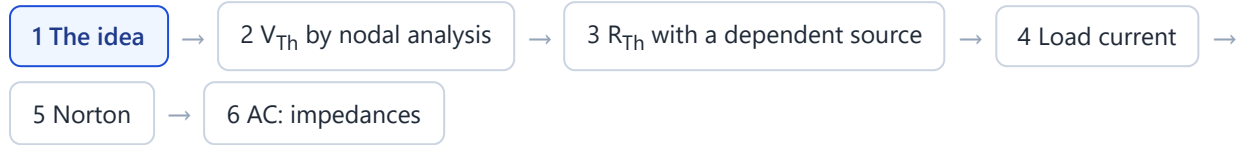
Self-test.

1. Response = algebraic sum of each independent source's response acting alone; the other independent sources are replaced by their internal resistances; dependent sources stay.
2. ∞ ; it becomes an **open** circuit.
3. 10 V alone: $V_A = 10/3$ V. 2 A alone: $V_A = 2 \times (4 \parallel 2) = 2 \times 4/3 = 8/3$ V. Sum = **6 V** ✓ (matches the node equation).
4. No. Superpose the current (or voltage), then compute $P = I^2R$.
5. **One**: only independent sources get their own case.
6. Superposition: $0 + 0 = \mathbf{0 \text{ A}}$. True: **5 mA**. Parallel ideal voltage sources cannot be switched off one at a time.

06 — Thévenin's and Norton's theorems

Class notes p.10–11, p.15, set 2 p.10–13 · MTE syllabus item 5 (DC and AC) · 1 board question (with a dependent source) + 1 textbook AC problem · needs 01, 03, 04, 05

Map



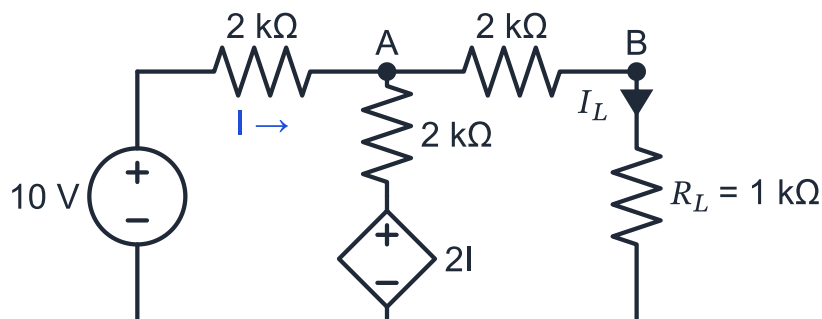
The question this file answers

- **Class p.11 · set 2 p.11** Determine the load current using Thévenin's theorem (10 V source, dependent source $2I$, $R_L = 1\text{ k}\Omega$).
- **textbook** AC: $10\angle 0^\circ\text{ V}$, $1\text{ k}\Omega$, $1\text{ }\mu\text{F}$ at $\omega = 1000\text{ rad/s}$. Find the Thévenin and Norton equivalents.

Build

CLASS P.11 · SET 2 P.11

Determine the load current I_L using Thévenin's theorem.



The board circuit. I is the current in the left $2\text{ k}\Omega$, flowing right. The dependent source is a CCVS of value $2I$, with $+$ at the top.

UNITS

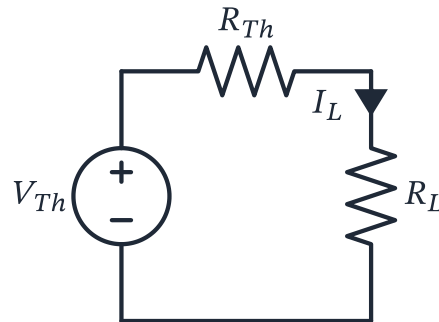
With $\text{k}\Omega$ and mA , " $2I$ " means $2\text{ k}\Omega \times I$ (01, step 3): $I = 1\text{ mA}$ gives 2 V . The set 2 working reads it differently and gets $V_{\text{Th}} = 5.0025\text{ V}$; the Note at the end of step 2 explains why both are right for their reading. The rest below is the standard method, with every number confirmed by computer.

GUESS FIRST

Without solving: if R_L changed from $1\text{ k}\Omega$ to $5\text{ k}\Omega$, would you want to re-solve the whole circuit? What would you rather have?

1 • The idea

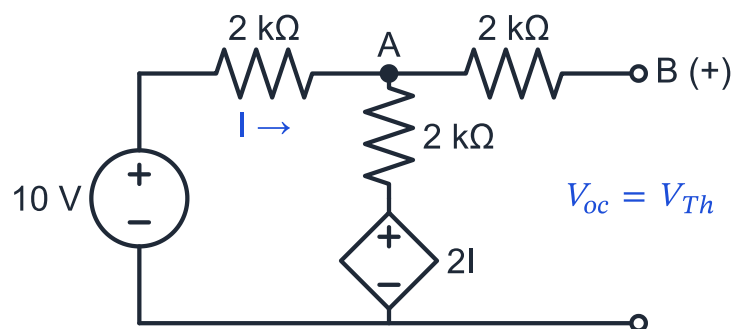
Any **linear** two-terminal network, however complicated, behaves at its terminals **exactly like one voltage source V_{Th} in series with one resistance R_{Th}** . Replace the whole network with that pair and the load is easy, for any R_L .



The Thévenin equivalent: $I_L = V_{Th} / (R_{Th} + R_L)$.

Procedure (class p.10): 1. Remove the load R_L (or Z_L). 2. Find V_{Th} , the **open-circuit** voltage across the load terminals. 3. Find R_{Th} across the same terminals. 4. Put R_L back on the equivalent: $I_L = V_{Th} / (R_{Th} + R_L)$.

2 • V_{Th} by nodal analysis (the professor's route)



Load removed. No current flows in the right $2\text{ k}\Omega$ (it leads to an open end),
so $V_{Th} = V_A$.

This is nodal analysis (04) with one unknown, V_A . Units: V, mA, $\text{k}\Omega$.

- Current in the left resistor, into A: $I = \frac{10 - V_A}{2}$
- Current in the middle branch, down: $I_1 = \frac{V_A - 2I}{2}$. No current in the right $2\text{ k}\Omega$, so KCL at A gives $I = I_1$
(set 2 p.11 writes exactly this).
- KCL: $\frac{10 - V_A}{2} = \frac{V_A - 2I}{2} \Rightarrow 10 - V_A = V_A - 2I$
- Substitute the controlling current, $2I = 2 \times \frac{10 - V_A}{2} = 10 - V_A$:

$$10 - V_A = V_A - (10 - V_A) \quad \Rightarrow \quad V_A = \frac{20}{3} \text{ V} \quad (\text{and } I = \frac{5}{3} \text{ mA})$$

$$V_{Th} = 20/3 \approx 6.67 \text{ V}$$

CHECK 1

Why does no current flow in the right-hand 2 k Ω here, and why does that make $V_{Th} = V_A$?

SET 2 P.12 GETS $V_{Th} = 5.0025 \text{ V}$. NOT AN ARITHMETIC ERROR: A UNIT READING

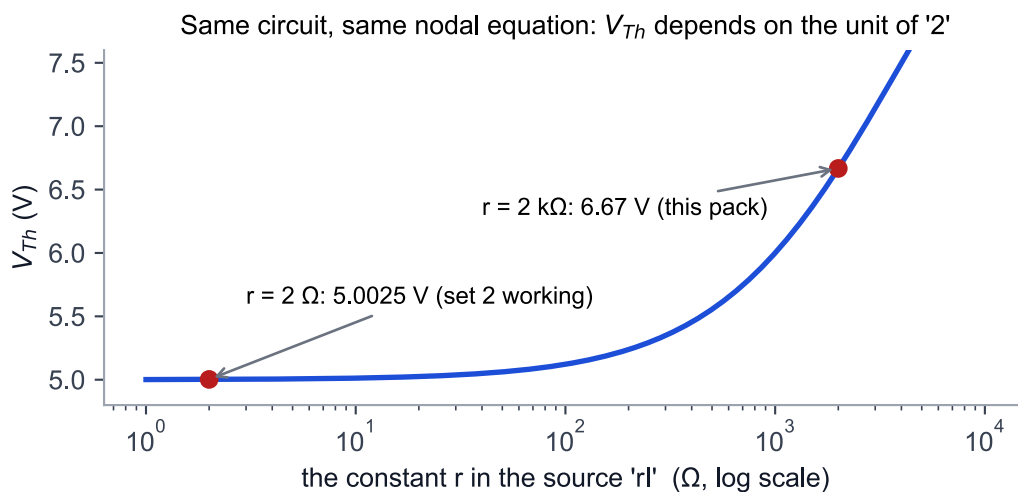
The set 2 working writes $I = \frac{10 - V_A}{2k}$ **in amperes** ($2k = 2000 \Omega$) and then uses the source as $2 \times I$ volts.

That treats the constant as **2 Ω** :

$$10 - V_A = V_A - 2 \cdot \frac{10 - V_A}{2000} \quad \Rightarrow \quad 10.01 = 2.001 V_A \quad \Rightarrow \quad V_A = 5.0025 \text{ V}$$

For that reading the arithmetic is right: a computer solve with a 2 Ω source gives 5.0025 V, $R_{Th} = 2.9995 \text{ k}\Omega$, $I_L = 1.2508 \text{ mA}$.

The pack keeps **2 k Ω** ($V_{Th} = 20/3 \text{ V}$, $I_L = 20/11 \text{ mA}$) because in a k Ω circuit written in mA, "2I" normally means 2 V per mA. And a 2 Ω source against 2 k Ω resistors changes V_{Th} by only 0.0025 V from the 5 V it would be without the source, so the question could not be showing off a dependent source. **Ask the professor which unit the "2" has.** In the exam, state your reading in one line ("2I with I in mA, i.e. 2 k Ω ") before solving.



V_{Th} from the same nodal equation for every value of the constant. The two readings are two points on one curve.

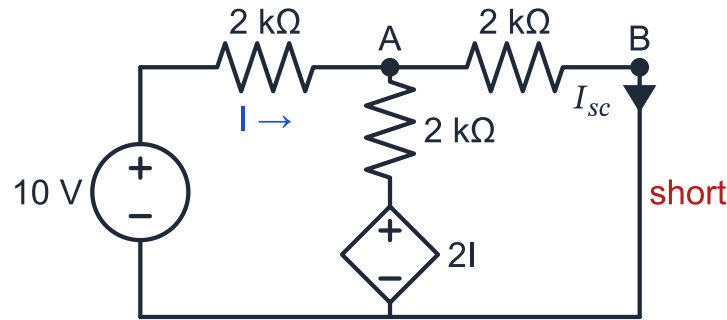
3 · R_{Th} when there is a dependent source

GUESS FIRST

Switch off the 10 V source and "look in" from B: 2 k Ω in series with $(2 \text{ k}\Omega \parallel 2 \text{ k}\Omega) = 3 \text{ k}\Omega$. What is wrong with doing that here?

The dependent source **stays on** (05), and it changes how current flows. So resistors alone don't give R_{Th} . Two correct methods; both give the same answer.

Method A: $R_{Th} = V_{oc} / I_{sc}$. Short the terminals and find the short-circuit current:

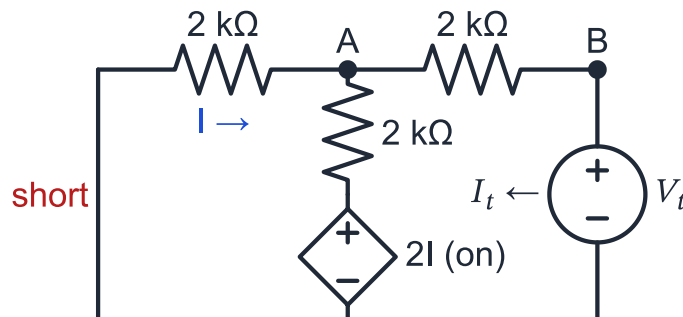


B shorted to the bottom rail. Now the right $2\text{ k}\Omega$ carries current too.

KCL at A: $I = \frac{V_A - 2I}{2} + \frac{V_A}{2} \Rightarrow V_A = 2I$. With $I = \frac{10 - 2I}{2}$: $I = 2.5\text{ mA}$, $V_A = 5\text{ V}$, so $I_{sc} = \frac{5}{2} = \mathbf{2.5\text{ mA}}$.

$$R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{20/3}{2.5} = 8/3 \approx 2.67\text{ k}\Omega$$

Method B: test source (the professor's method, class p.10: $R_{Th} = V_{dc} / I_{dc}$). Switch off the **independent** source only, keep $2I$, apply a test voltage V_t at the terminals and find the current I_t it drives:



10 V replaced by a short; the dependent source is still on; the test source V_t is at B.

With the 10 V shorted, the left $2\text{ k}\Omega$ runs from ground to A, so $I = \frac{0 - V_A}{2} = -\frac{V_A}{2}$. Currents leaving A downwards:

- through the left $2\text{ k}\Omega$: $\frac{V_A}{2}$
- through the middle branch: $\frac{V_A - 2I}{2} = \frac{V_A + V_A}{2} = V_A$

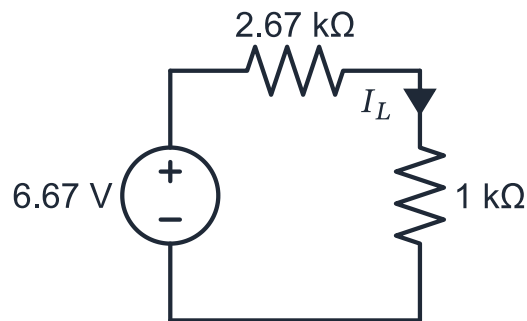
Total = $1.5 V_A$, so A looks like $\frac{1}{1.5} = \frac{2}{3}\text{ k}\Omega$ to ground. Then $R_{Th} = 2 + \frac{2}{3} = \mathbf{8/3\text{ k}\Omega}$ ✓, the same as method A.

The 3 kΩ from the Guess box is **wrong**: it ignores that the dependent source doubles the middle branch's current.

CHECK 2

(a) In method B, why is I negative? (b) If the circuit had **no** independent source at all, which method would still work, and why?

4 • The load current



The whole left side, replaced by its Thévenin equivalent.

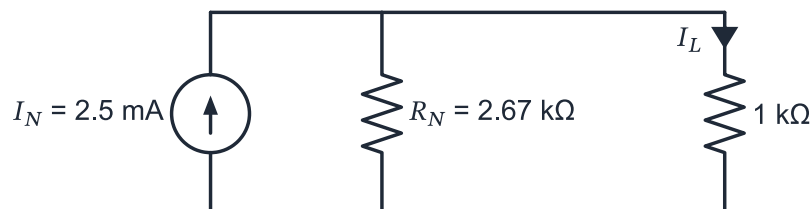
$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{20/3}{8/3 + 1} = \frac{20}{11} \approx 1.82 \text{ mA}$$

Confirmed by solving the full original circuit with R_L in place: **1.818 mA**. For $R_L = 5 \text{ k}\Omega$ it is now one line: $6.67 / 7.67 = 0.87 \text{ mA}$.

5 • Norton: the same network as a current source

Norton's equivalent is a **current source I_N in parallel with R_N** (class p.15). By source transformation (03, step 4):

$$I_N = I_{sc} = \frac{V_{Th}}{R_{Th}} \quad R_N = R_{Th}$$



Norton equivalent of the same network: 2.5 mA in parallel with 2.67 kΩ.

Current division: $I_L = 2.5 \times \frac{8/3}{8/3 + 1} = \mathbf{20/11 \text{ mA}}$ ✓. Same answer.

Norton procedure (class p.15): (i) remove the load; (ii) find R_N , exactly as for R_{Th} ; (iii) short the load terminals and find the short-circuit current I_N ; (iv) draw $I_N \parallel R_N$ with the load.

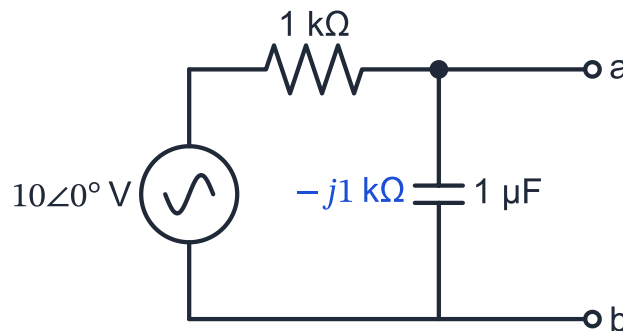
CHECK 3

A network has $V_{Th} = 12 \text{ V}$ and $R_{Th} = 4 \text{ k}\Omega$. Give its Norton equivalent.

6 • Thévenin and Norton with AC sources

TEXTBOOK

A source $10\angle 0^\circ \text{ V}$ (peak) at $\omega = 1000 \text{ rad/s}$ drives $1 \text{ k}\Omega$ in series, with $1 \mu\text{F}$ across the output a–b. Find the Thévenin and Norton equivalents at a–b.



$$\text{The capacitor's impedance at } \omega = 1000: Z_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 1000 \cdot 10^{-6}} = -j1000 \Omega.$$

GUESS FIRST

Is the open-circuit voltage 10 V, less, or more? Is it in phase with the source?

Nothing new in the method: the same four steps, with every resistance replaced by an **impedance** Z ($R \rightarrow R$, $L \rightarrow j\omega L$, $C \rightarrow \frac{1}{j\omega C}$) and every value a **phasor** (a complex number carrying amplitude and phase). Work in $\text{k}\Omega$:

$$Z_C = -j1 \text{ k}\Omega.$$

V_{Th} : open circuit, so the capacitor and the $1 \text{ k}\Omega$ form a voltage divider (02):

$$V_{Th} = 10 \cdot \frac{-j1}{1 - j1} = 10 \cdot \frac{-j(1 + j)}{2} = 5 - j5 = 7.07\angle -45^\circ \text{ V}$$

Z_{Th} : source $\rightarrow$ short, look in from a–b: $1 \text{ k}\Omega$ in parallel with $-j1 \text{ k}\Omega$:

$$Z_{Th} = \frac{1 \cdot (-j1)}{1 - j1} = 0.5 - j0.5 \text{ k}\Omega = 500 - j500 \Omega$$

Norton: $I_N = \frac{V_{Th}}{Z_{Th}} = \frac{5 - j5}{0.5 - j0.5} = 10\angle 0^\circ \text{ mA}$. Check: shorting a–b shorts the capacitor, leaving 10 V across 1 k Ω : 10 mA $\checkmark$.

The Guess answer: **less** (7.07 V) and **lagging** by 45° . Unlike the resistor divider in 02, an impedance divider changes the phase.

The Thévenin equivalent is $7.07\angle -45^\circ \text{ V}$ in series with $500 - j500 \Omega$ (a resistor with a capacitive part). 07 §4 finds the load that takes the most power from it.

CHECK 4

Same circuit at $\omega = 2000 \text{ rad/s}$. Find Z_C and V_{Th} .

Exam form

Thévenin's theorem. Any linear bilateral network seen from two terminals can be replaced by a voltage source V_{Th} (the open-circuit voltage at those terminals) in series with R_{Th} (the resistance seen at the terminals with all independent sources switched off).

Norton's theorem. The same network can be replaced by a current source I_N (the short-circuit current) in parallel with $R_N = R_{Th}$.

Link: $V_{Th} = I_N \cdot R_{Th}$.

R_{Th} with dependent sources: keep them on; use V_{oc}/I_{sc} or apply a test source: $R_{Th} = V_t/I_t$.

AC: the same, with impedances ($Z_L = j\omega L$, $Z_C = 1/j\omega C$) and phasors: V_{Th} and Z_{Th} are complex; $I_N = V_{Th}/Z_{Th}$.

Traps

- Switching off the dependent source when finding R_{Th} .
- Using the test-source method but forgetting to switch off the **independent** sources (the class p.10 sketch still shows V_{in} ; it must be zeroed).
- Measuring V_{Th} with the load still connected.
- Norton: I_N is the **short-circuit** current, not the current in the original load.
- Mixing A and mA in one nodal equation (set 2 p.12): decide the unit of every dependent-source constant first.
- AC: adding impedance magnitudes instead of the complex numbers (1 k Ω and $-j1 \text{ k}\Omega$ in series is 1.414 k Ω in size, not 2 k Ω).

Self-test

1. State Thévenin's theorem in two lines.
2. Which two measurements, both at the terminals, give V_{Th} and R_{Th} for any linear network?
3. For the board circuit, what would R_{Th} be if the dependent source were replaced by a plain wire?
4. Find I_L for the board circuit if $R_L = 8/3 \text{ k}\Omega$. What is special about that value? (Hint: 07.)
5. Convert $V_{Th} = 6 \text{ V}$, $R_{Th} = 3 \text{ k}\Omega$ into Norton form.

Answers

Check 1. The right $2\text{ k}\Omega$ ends at an open terminal, so there is no path: $I = 0$, voltage drop = 0, so $V_B = V_A$.

Check 2. (a) With the 10 V shorted, current flows from ground *into* A through the left resistor, which is opposite to the defined direction of I . (b) Only the test-source method: $V_{oc} = 0$ and $I_{sc} = 0$, so V_{oc}/I_{sc} is $0/0$.

Check 3. $I_N = 12/4 = \mathbf{3\text{ mA}}$ in parallel with $\mathbf{4\text{ k}\Omega}$.

Check 4. $Z_C = 1/(j \cdot 2000 \cdot 10^{-6}) = \mathbf{-j500\ \Omega}$. $V_{Th} = 10 \times \frac{-j0.5}{1 - j0.5} = \mathbf{4.47\angle -63.4^\circ\text{ V}}$ ($= 2 - j4\text{ V}$).

Self-test.

1. See Exam form.
2. The open-circuit voltage ($= V_{Th}$) and the short-circuit current (I_{sc}); $R_{Th} = V_{oc}/I_{sc}$.
3. The middle branch becomes a plain $2\text{ k}\Omega$: $R_{Th} = 2 + (2 \parallel 2) = \mathbf{3\text{ k}\Omega}$.
4. $\frac{20/3}{16/3} = \mathbf{1.25\text{ mA}}$. $R_L = R_{Th}$: this load receives the maximum possible power.
5. $\mathbf{2\text{ mA}}$ in parallel with $\mathbf{3\text{ k}\Omega}$.

07 — Maximum power transfer

Class notes p.13–14, p.16–17, set 2 p.14 · MTE syllabus item 5 (DC and AC) · 2 board question sets + 1 textbook AC problem · needs 06

Map

1 Power in the load

→

2 The condition $R_L = R_{Th}$

→

3 P_{max} and the 50 % efficiency

→

4 AC loads: conjugate match

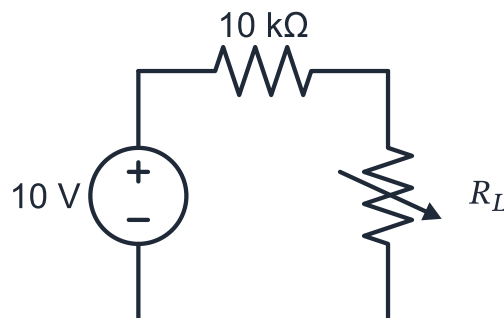
The questions this file answers

- **Class p.16–17** 10 V source, 10 k Ω , variable R_L : for what R_L is power maximum? Total power from the source? Power in R_{Th} ? Efficiency? Voltage across the load? How do the load's voltage and power compare with the source's?
- **Class p.17** 10 V source, 5 k Ω : find R for maximum power, the maximum power, and the efficiency.

Build

CLASS P.16

For what value of load resistance will maximum power be delivered? Then: determine the total power delivered by the source.



The board circuit: a 10 V source with 10 k Ω in series, driving a variable load.

GUESS FIRST

Make R_L tiny, then huge. What happens to the load's power at each extreme? So where must the maximum be?

1 · Power in the load

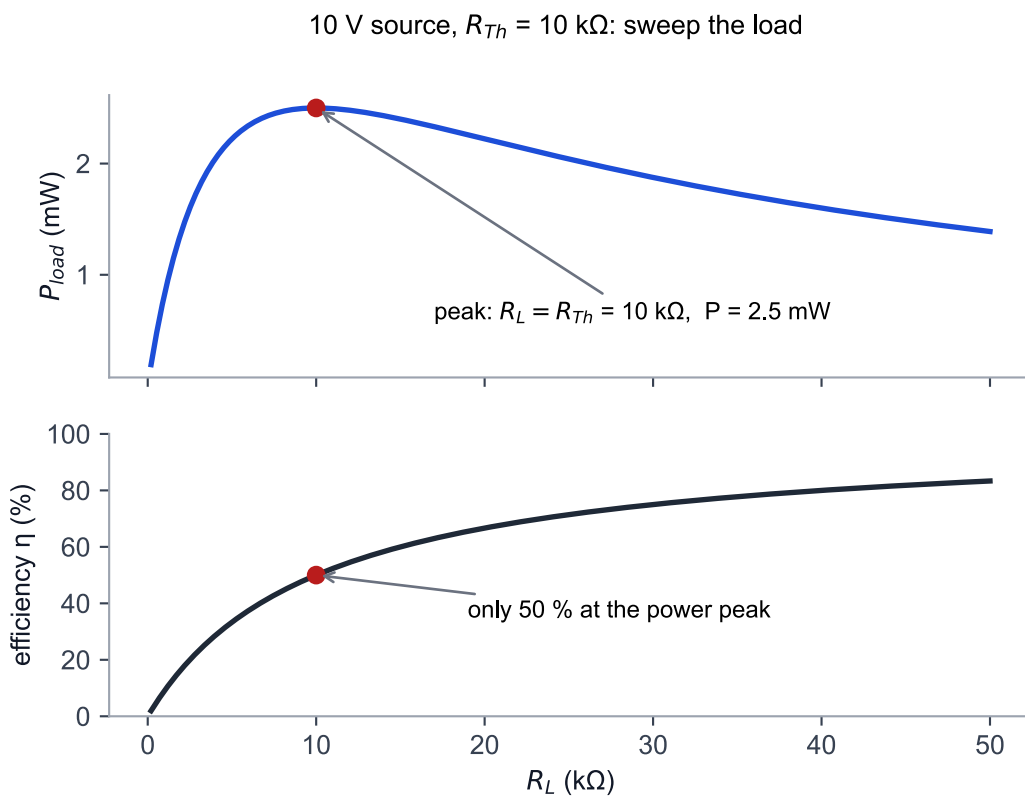
From 06, every linear source network is V_{Th} in series with R_{Th} , so this circuit is already in that form. The load current and power:

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} \quad P_L = I_L^2 R_L = \frac{V_{Th}^2 R_L}{(R_{Th} + R_L)^2}$$

CLASS P.13

The first attempt on the board used $P_L = V_{Th}^2 / (R_{Th} + R_L)$, which is wrong; it was crossed out and redone with $P_L = I_L^2 R_L$ (starred). Use the starred one.

The Guess answer: tiny R_L gives big current but almost no voltage, so $P \approx 0$. Huge R_L gives big voltage but almost no current, so $P \approx 0$. The power must peak somewhere in between:



Top: the load's power peaks exactly at $R_L = R_{Th}$. Bottom: efficiency keeps rising with R_L and is only 50 % at that peak.

2 · The condition (class p.13–14 derivation)

Set $\frac{dP_L}{dR_L} = 0$. Quotient rule:

$$\frac{dP_L}{dR_L} = V_{Th}^2 \cdot \frac{(R_{Th} + R_L)^2 - R_L \cdot 2(R_{Th} + R_L)}{(R_{Th} + R_L)^4} = 0$$

The numerator must be zero:

$$(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0 \Rightarrow R_{Th}^2 + 2R_{Th}R_L + R_L^2 - 2R_{Th}R_L - 2R_L^2 = 0 \Rightarrow R_{Th}^2 = R_L^2$$

$$R_L = R_{Th}$$

(The negative root is not a resistance. The curve above shows it is a maximum, not a minimum.)

Answer, part 1: $R_L = R_{Th} = 10 \text{ k}\Omega$.

3 · P_{max} , source power, efficiency

Put $R_L = R_{Th}$ into P_L :

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

For the board circuit, at $R_L = 10 \text{ k}\Omega$:

Quantity	Working	Value
Current	10 V / 20 k Ω	0.5 mA
Total power from the source	$V \cdot I = 10 \times 0.5 \text{ mA}$	5 mW
Power in the load	$I^2 R_L = (0.5 \text{ mA})^2 \times 10 \text{ k}\Omega = V_{Th}^2 / 4R_{Th}$	2.5 mW
Power in R_{Th}	same current, same resistance	2.5 mW
Efficiency η	$P_{load} / P_{total} = 2.5 / 5$	50 %
Voltage across the load	$0.5 \text{ mA} \times 10 \text{ k}\Omega$	5 V = $V_s/2$
Load vs source		$V_L = V_s/2, P_L = P_s/2$

CLASS P.16

The notes read " $I \times V = 0.5 \text{ mW}$ " with no visible decimal point. The value is **5 mW**. The 50 % efficiency on p.17 is right either way.

Maximum power is not maximum efficiency. $\eta = \frac{R_L}{R_{Th} + R_L}$ keeps rising as R_L grows (bottom plot), while P_L falls. So:

- **communication / signal circuits** match $R_L = R_{Th}$: a weak signal needs every bit of power delivered, and 50 % loss is acceptable;
- **power systems** keep $R_L \gg R_{Th}$: wasting half the generated power would be unacceptable.

The 50 % refers to the Thévenin equivalent. The power actually lost inside the original, full network can differ.

CHECK 1

- (a) Why is exactly half the power lost at the match? Use the fact that R_{Th} and R_L carry the same current.
- (b) The class p.17 question: 10 V with 5 k Ω . Find R , P_{max} , η .

4 • AC loads: the conjugate match

PREDICT

A source has $Z_{Th} = 3 + j4 \Omega$. Its reactance $j4 \Omega$ doesn't absorb power, but it does limit the current. What reactance should the load add?

For impedances (class p.16; set 2 p.14: " $Z_L = Z_{Th}^*$, $*$ = complex conjugate, $i \rightarrow -i$):

$$Z_L = Z_{Th}^* \quad (\text{the complex conjugate: } j \rightarrow -j)$$

- $R_L = R_{Th}$: the resistance match, as before.
- $X_L = -X_{Th}$: the load's reactance **cancels** the source's, so the loop is purely resistive and the current is as large as possible.

For the Predict: $Z_L = 3 - j4 \Omega$. Then $P_{max} = \frac{|V_{Th}|^2}{4R_{Th}}$, with V_{Th} as an **rms** value (with a peak value it is $\frac{|V_{Th}|^2}{8R_{Th}}$).

TEXTBOOK

The AC network of 06 §6 has $V_{Th} = 7.07 \angle -45^\circ$ V (peak) and $Z_{Th} = 500 - j500 \Omega$. Find the load for maximum power, and that power.

- **Load:** $Z_L = Z_{Th}^* = 500 + j500 \Omega$ (500 Ω with an inductive reactance of 500 Ω ; at $\omega = 1000$ that is $L = 0.5$ H).
- **Loop:** $Z_{Th} + Z_L = 1000 \Omega$, purely resistive: the reactances cancel. $I = \frac{7.07}{1000} = 7.07$ mA peak = 5 mA rms.
- **Power:** $P = I_{rms}^2 R_L = (5 \text{ mA})^2 \times 500 \Omega = 12.5 \text{ mW}$. Same by formula: $V_{rms} = 5 \text{ V}$, $\frac{25}{4 \times 500} = 12.5 \text{ mW}$.

A computer sweep over every load $R + jX$ gives its peak at exactly $R = 500 \Omega$, $X = +500 \Omega$, 12.5 mW.

Exam form

Statement. A linear network delivers maximum power to a resistive load when $R_L = R_{Th}$. The maximum power is $P_{max} = V_{Th}^2 / (4R_{Th})$. For an AC network with a complex load, the condition is $Z_L = Z_{Th}^*$.

Derivation. $P_L = V_{Th}^2 R_L / (R_{Th} + R_L)^2$; set $dP_L / dR_L = 0 \Rightarrow R_L = R_{Th}$ (step 2).

At the match. $\eta = 50 \%$, $V_L = V_{Th} / 2$, $P_L = P_S / 2$.

Traps

- Using $P = V_{Th}^2 / (R_{Th} + R_L)$ (the crossed-out attempt on class p.13).
- Saying "maximum power means maximum efficiency".
- In the AC case, matching $Z_L = Z_{Th}$ instead of the **conjugate**.
- Forgetting to find R_{Th} first when the source network is more than one resistor: use 06.

Self-test

1. Derive $R_L = R_{Th}$ (three lines).
2. $V_{Th} = 12\text{ V}$, $R_{Th} = 3\text{ k}\Omega$. Find P_{max} and the source power at that point.
3. Why do power companies not operate at maximum power transfer?
4. $Z_{Th} = 50 - j20\ \Omega$. Load for maximum power? If $V_{Th} = 10\text{ V rms}$, what is P_{max} ?
5. In the 06 board circuit ($V_{Th} = 20/3\text{ V}$, $R_{Th} = 8/3\text{ k}\Omega$), what R_L gets maximum power, and how much?

Answers

Check 1. (a) At the match, $R_{Th} = R_L$ and both carry the same current, so $I^2 R_{Th} = I^2 R_L$: equal power in each, i.e. half of the total. (b) $R = 5 \text{ k}\Omega$; $P_{max} = 10^2 / (4 \times 5 \text{ k}\Omega) = 5 \text{ mW}$; $\eta = 50 \%$ (the source supplies $10 \text{ V} \times 1 \text{ mA} = 10 \text{ mW}$).

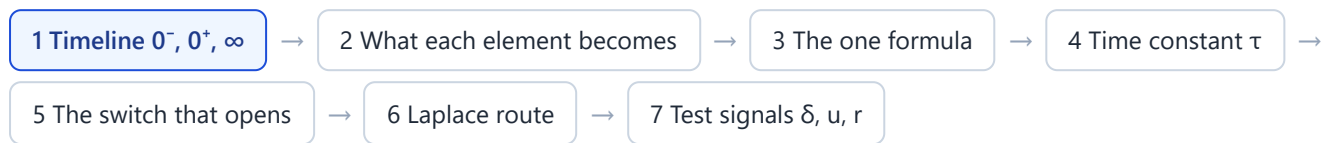
Self-test.

1. $P_L = V_{Th}^2 R_L / (R_{Th} + R_L)^2$; the numerator of the derivative is $(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0$; this gives $R_{Th}^2 = R_L^2$, so $R_L = R_{Th}$.
2. $P_{max} = 144/12\text{k} = 12 \text{ mW}$; $I = 12/6\text{k} = 2 \text{ mA}$; $P_s = 24 \text{ mW}$.
3. Efficiency would be only 50 %: half of the generated power lost in the source and lines.
4. $Z_L = 50 + j20 \Omega$; $P_{max} = 100 / (4 \times 50) = 0.5 \text{ W}$.
5. $R_L = 8/3 \text{ k}\Omega \approx 2.67 \text{ k}\Omega$; $P_{max} = (20/3)^2 / (4 \times 8/3) = 4.17 \text{ mW}$.

08 — First-order transients

Class notes p.19–23, set 2 p.15–22 · MTE syllabus item 6 (step-wise, Laplace, test signals) · 4 board questions + 2 textbook · needs 01 (what can't jump), 06 (R_{Th} for the time constant)

Map



The questions this file answers

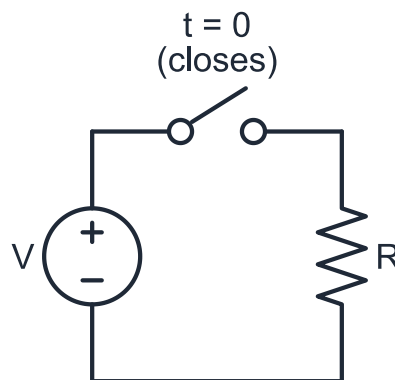
- **Class p.19–20** V , a switch that closes at $t = 0$, R : find I at 0^- , 0^+ , ∞ .
- **Class p.21** V , switch, R , L in series; the switch closes at $t = 0$: find I_R and I_L at 0^- , 0^+ , ∞ .
- **Set 2 p.20–21** V , switch (closes at $t = 0$), R , C in series: find v_C , v_R and I at 0^- , 0^+ , ∞ .
- **Class p.22–23 · set 2 p.18–19** 10 V , switch (opens at $t = 0$), 1 $k\Omega$, 1 mH : find I_R and I_L at 0^- and 0^+ .
- **textbook** The class RL circuit solved by Laplace; RC responses to $\delta(t)$, $u(t)$, $r(t)$.

Build

1 · The timeline

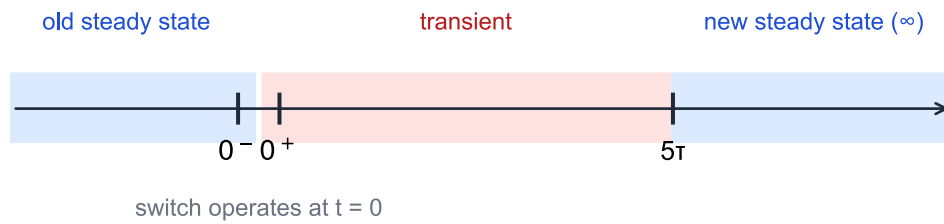
CLASS P.19–20 · SET 2 P.15–16

V , a switch that closes at $t = 0$, and R . Find $I(0^-)$, $I(0^+)$ and $I(\infty)$.



GUESS FIRST

Is there any "transient" at all in this circuit, i.e. any gradual change?



Before the switch: the old steady state. From 0^+ to about 5τ : the transient. After that: the new steady state.

- $t = 0^-$: the instant **just before** the switch operates. Old steady state.
- $t = 0^+$: the instant **just after**.
- $t = \infty$: long after (in practice about 5τ): new steady state.

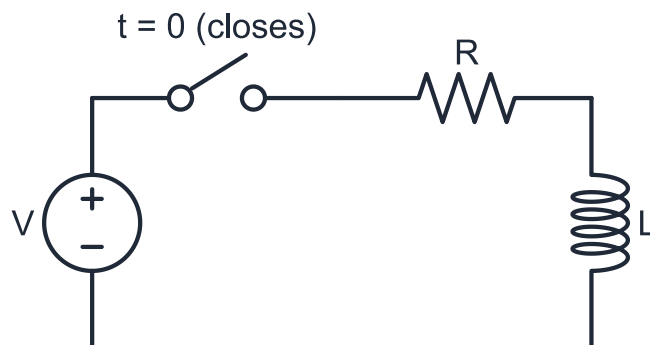
For the resistor-only circuit: $I(0^-) = 0$ (switch open), $I(0^+) = V/R$, $I(\infty) = V/R$. The current jumps straight to its final value, because a resistor has no memory (class p.20: "a resistor allows sudden change").

So the Guess answer: **no transient**. A transient needs an element that stores energy, an L or a C.

2 · What each element becomes at 0^+ and at ∞

CLASS P.21 · SET 2 P.17

V , switch, R and L in series; the switch closes at $t = 0$. Find I_R and I_L at 0^- , 0^+ and ∞ .

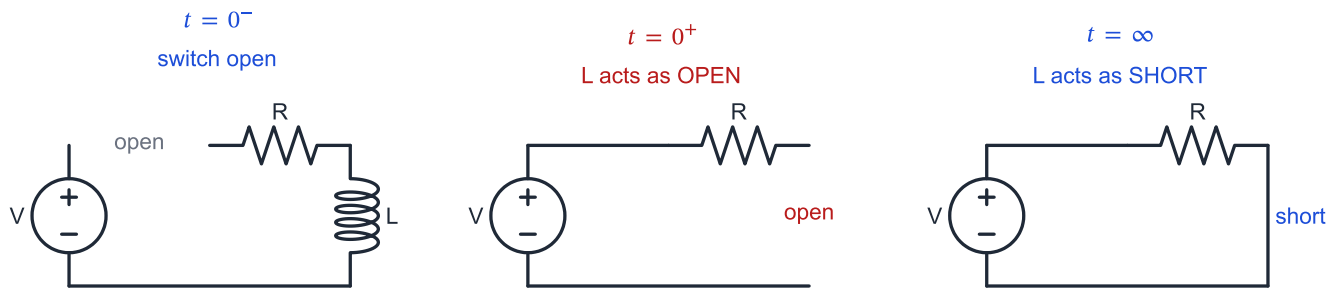


GUESS FIRST

The switch has just closed. The source is pushing. Why can't the current be V/R at $t = 0^+$?

Two rules do all the work (from 01, step 5):

Element	At 0^+ : it keeps its 0^- value	At ∞ (DC steady state)
Inductor	a current source of value $i_L(0^-)$; an open circuit if that value is 0	short circuit
Capacitor	a voltage source of value $v_C(0^-)$; a short circuit if that value is 0	open circuit
Resistor	no memory: whatever the rest of the circuit forces	same



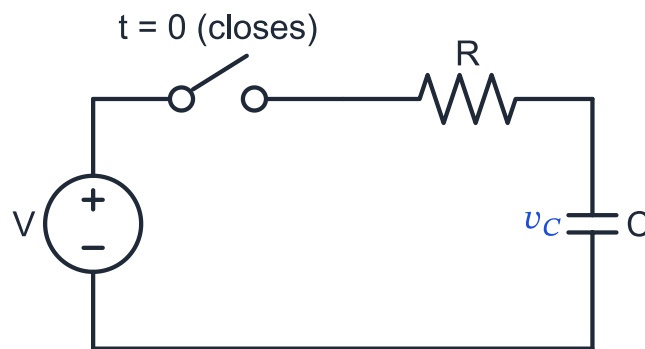
The three snapshots of the class p.21 circuit.

- 0^- : switch open, so $I_R(0^-) = I_L(0^-) = 0$.
- 0^+ : i_L can't jump, so $I_L(0^+) = 0$ and the inductor acts as an **open**. In series, $I_R(0^+) = 0$.
- ∞ : the inductor is a **short**, so $I(\infty) = V/R$.

The current must travel from 0 to V/R . How it travels is step 3.

SET 2 P.20-21

V , a switch that closes at $t = 0$, R and C in series (C uncharged). Find v_C , v_R and I at 0^- , 0^+ and ∞ .



The mirror image of the inductor:

- 0^- : switch open, steady state: $v_C(0^-) = 0$, $v_R(0^-) = 0$, $I = 0$.
- 0^+ : **$v_C(0^+) = v_C(0^-) = 0$** . The class writes it as the rule to remember: " $v_C(0^-) = v_C(0^+)$, always". So the capacitor acts as a **short**, the whole V is across R : $v_R(0^+) = V$, **$I(0^+) = V/R$** . The current jumps; the capacitor voltage does not.
- ∞ : $I = C \cdot \frac{dv_C}{dt} = 0$ because v_C is constant, so the capacitor is an **open** (set 2 p.21): **$I(\infty) = 0$, $v_C(\infty) = V$, $v_R(\infty) = 0$** .

CHECK 1

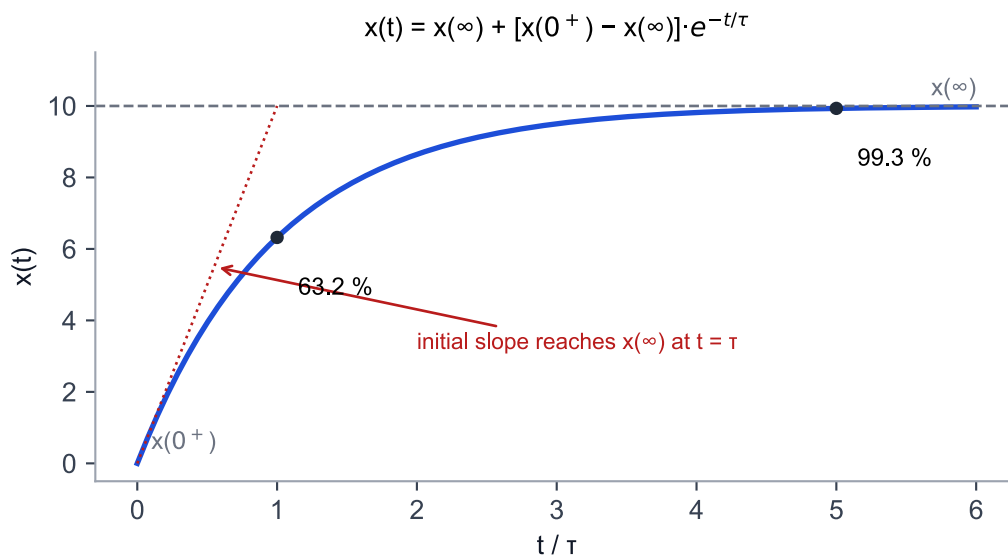
Same RC circuit, but C was charged to 4 V (+ at the top) before $t = 0$, with $V = 10$ V and $R = 1$ k Ω . Find $I(0^+)$ and $I(\infty)$.

3 · The one formula for every first-order circuit

Every circuit with **one** L or **one** C (after reduction) moves from its 0^+ value to its ∞ value **exponentially**:

$$x(t) = x(\infty) + [x(0^+) - x(\infty)] \cdot e^{-t/\tau}$$

The class p.22 box: valid for **RL ✓ and RC ✓**, not for **RLC X or LC X** (those are second order: they can oscillate).



After one τ the change is 63.2 % complete; after 5τ , 99.3 %. The initial slope would reach $x(\infty)$ in exactly one τ .

So a first-order question is always **three numbers**: $x(0^+)$, $x(\infty)$, τ . Then write the formula.

4 • The time constant τ

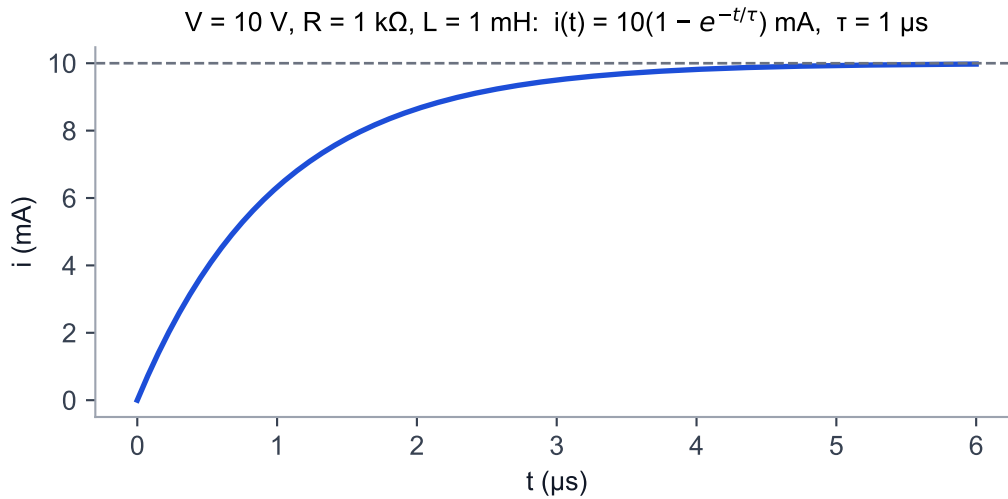
$$\tau = \frac{L}{R} \quad (\text{RL}) \quad \tau = RC \quad (\text{RC})$$

R is the Thévenin resistance seen by the L or C (06), with independent sources switched off. It is the plain R only when the L or C sees a single resistor.

Class p.21 circuit with the p.22 values ($V = 10 \text{ V}$, $R = 1 \text{ k}\Omega$, $L = 1 \text{ mH}$): $x(0^+) = 0$, $x(\infty) = 10 \text{ mA}$, $\tau = \frac{1 \text{ mH}}{1 \text{ k}\Omega} = 1$

μs :

$$i(t) = 10(1 - e^{-t/1 \mu\text{s}}) \text{ mA}$$



Confirmed by simulating the circuit step by step: at $t = 3 \mu\text{s}$ both give 9.50 mA.

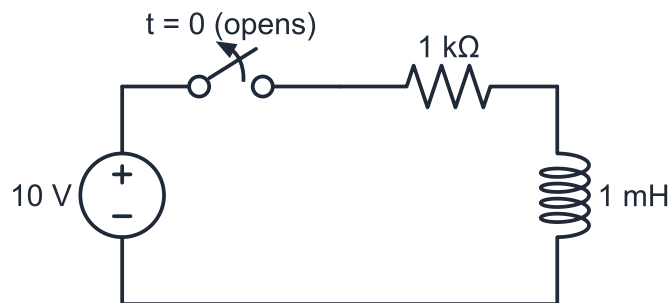
CHECK 2

An RC circuit: $R = 2 \text{ k}\Omega, C = 5 \mu\text{F}$. Find τ , and the time after which it is "settled" (99 %).

5 • The switch that opens

CLASS P.22–23 • SET 2 P.18–19

10 V, a switch that opens at $t = 0$, $1 \text{ k}\Omega$ and 1 mH in series. Find I_R and I_L at 0^- and 0^+ .



GUESS FIRST

At 0^- the loop is carrying current. At 0^+ the switch is open. What must the inductor do, and where can its current go?

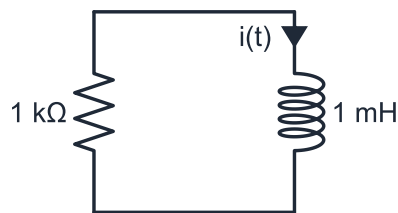
- 0^- : switch closed, steady state, $L = \text{short}$: $I_L(0^-) = 10 \text{ V} / 1 \text{ k}\Omega = 10 \text{ mA} = I_R(0^-)$.
- 0^+ : the inductor current can't jump, so $I_L(0^+) = 10 \text{ mA} \checkmark$ (the professor's point).

READ THIS BEFORE WRITING $I_R(0^+) = 0$

Both sets of notes give $I_R(0^+) = 0$ alongside $I_L(0^+) = 10 \text{ mA}$ (set 2 p.19 draws the inductor as a 10 mA source). R and L are **in the same series loop**, so they must carry the **same** current. Both statements can't hold. With the switch open there is no path at all for the inductor's 10 mA. In an ideal circuit, $di/dt \rightarrow \infty$ and the inductor voltage spikes. In a real one, the stored energy $\frac{1}{2}Li^2 = \frac{1}{2} \times 1 \text{ mH} \times (10 \text{ mA})^2 = 50 \text{ nJ}$ arcs across the opening contacts.

Ask the professor which path the current takes after $t = 0$. Usually the full question has the switch move the R–L onto its own loop, or has a resistor across the switch.

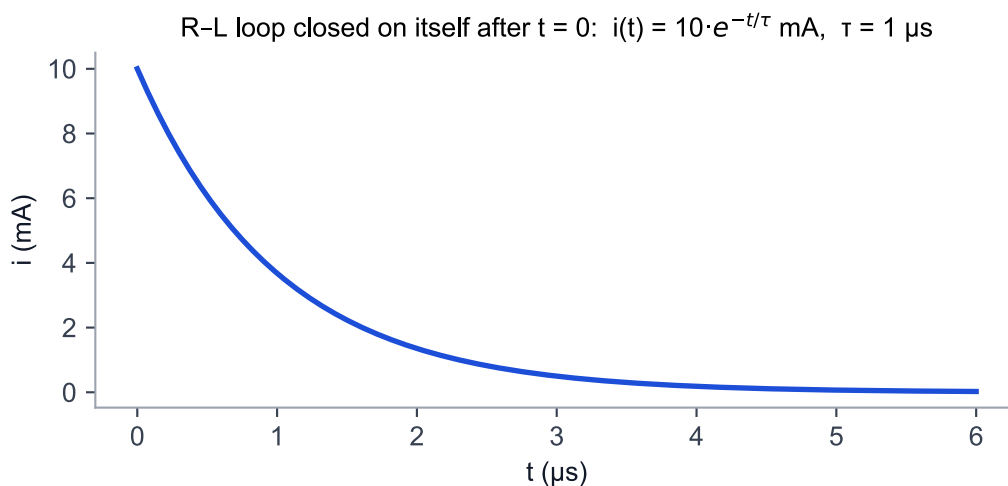
If the R–L loop is closed on itself after $t = 0$ (the usual version of this question):



Illustration, not the board circuit: the R–L loop with the source gone.

$x(0^+) = 10 \text{ mA}$, $x(\infty) = 0$, $\tau = L/R = 1 \mu\text{s}$:

$$i(t) = 10 \cdot e^{-t/1 \mu\text{s}} \text{ mA}$$



6 · The Laplace route

TEXTBOOK

Solve the class p.21 circuit (10 V switched onto 1 kΩ and 1 mH at $t = 0$) by Laplace transform.

GUESS FIRST

The answer is already known from step 4. What should the Laplace working give back, and at which step does the "1 - e" shape appear?

The step-wise method needs the exponential shape to be known in advance. Laplace **derives** it: it turns d/dt into multiplication by s, so the circuit's differential equation becomes algebra.

Pairs you need (the 06-KB table; the full list is in step 7):

f(t)	F(s)
$\delta(t)$	1
u(t) (a DC source switched on at 0)	$\frac{1}{s}$
t·u(t)	$\frac{1}{s^2}$
$e^{-at}u(t)$	$\frac{1}{s + a}$
$\frac{df}{dt}$	$sF(s) - f(0^-)$

Elements in the s-domain. $R \rightarrow R$. $L \rightarrow sL$. $C \rightarrow \frac{1}{sC}$. Energy stored at $t = 0^-$ becomes a source in series:



Current enters the top terminal. The inductor's initial current appears as a voltage source $L \cdot i_L(0^-)$; the capacitor's initial voltage as $v_C(0^-)/s$.

The class circuit. $i_L(0^-) = 0$, so no initial source. The 10 V switched on at $t = 0$ is $10 \cdot u(t) \rightarrow \frac{10}{s}$. KVL in s:

$$\frac{10}{s} = (R + sL) \cdot I(s) \quad \Rightarrow \quad I(s) = \frac{10}{s(1000 + 0.001s)} = \frac{10^4}{s(s + 10^6)}$$

Partial fractions: $\frac{10^4}{s(s + 10^6)} = \frac{A}{s} + \frac{B}{s + 10^6}$, with $A = \frac{10^4}{10^6} = 0.01$ and $B = -0.01$:

$$I(s) = \frac{0.01}{s} - \frac{0.01}{s + 10^6} \quad \Rightarrow \quad i(t) = 0.01(1 - e^{-10^6 t}) \text{ A} = 10(1 - e^{-t/1 \mu\text{s}}) \text{ mA}$$

The same answer as step 4. The "1 - e" comes from the two partial fractions: $1/s$ gives the final value, $1/(s + 1/\tau)$ gives the decaying part. The pole at $s = -10^6 = -\frac{1}{\tau}$ is the time constant.

With stored energy. The step 5 decay (R-L loop, $i_L(0^-) = 10$ mA, no source): KVL with the initial-current source, $0 = (R + sL)I(s) - L \cdot i_L(0^-)$, so

$$I(s) = \frac{i_L(0^-)}{s + R/L} \Rightarrow i(t) = 10e^{-t/1 \mu s} \text{ mA}$$

Quick checks without inverting (final- and initial-value theorems): $i(\infty) = \lim_{s \rightarrow 0} sI(s) = \frac{10^4}{10^6} = 10 \text{ mA} \checkmark$; $i(0^+) = \lim_{s \rightarrow \infty} sI(s) = 0 \checkmark$. The final-value theorem only holds if the response settles (no poles on the $j\omega$ axis or to the right).

CHECK 3

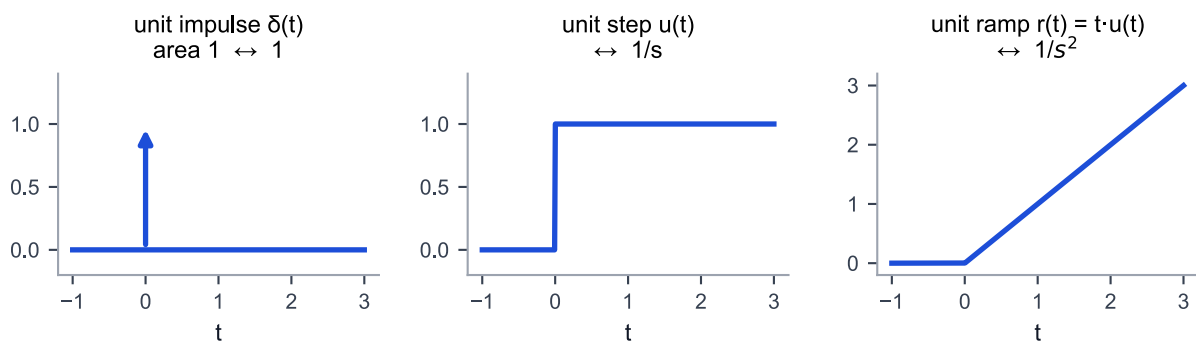
Write $I(s)$ for 10 V switched onto $R = 2 \text{ k}\Omega$ and $C = 1 \mu\text{F}$ in series (C uncharged), and invert it.

7 · Standard test signals: $\delta(t)$, $u(t)$, $r(t)$

PREDICT

A step is the integral of an impulse. If you know how a circuit responds to an impulse, how could you get its step response without solving again?

The class list (set 2 p.22), with its everyday examples:



The three test signals and their Laplace transforms.

Signal	Definition	$\mathcal{L}$	Class example	Tests...
unit impulse $\delta(t)$	zero except at $t = 0$; area $\int \delta(t) dt = 1$	1	a kitchen lighter's spark, a bike's spark plug	the response to a sudden kick
unit step $u(t)$	0 for $t < 0$, 1 for $t > 0$	$\frac{1}{s}$	a switch closing	the response to a sudden constant input
unit ramp $r(t)$	$t \cdot u(t)$	$\frac{1}{s^2}$	pressing an accelerator	tracking a steadily rising input: the "toughest" test signal (class)

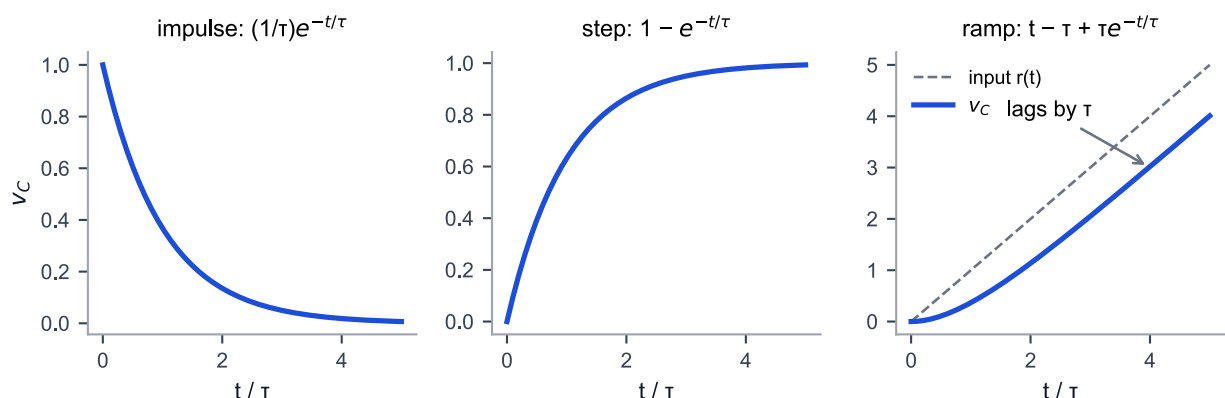
They are linked: $\delta(t) = \frac{du}{dt}$ and $u(t) = \frac{dr}{dt}$. So in a linear circuit **the step response is the integral of the impulse response**, and the ramp response is the integral of the step response. The Predict answer: integrate.

TEXTBOOK

RC circuit: $R = 1 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$ ($\tau = RC = 1 \text{ ms}$), initially uncharged, output v_C . Find $v_C(t)$ for a unit impulse, a unit step and a unit ramp at the input.

Voltage divider in s : $V_C(s) = V_{in}(s) \cdot \frac{1/sC}{R + 1/sC} = V_{in}(s) \cdot \frac{1}{1 + s\tau}$.

Input	$V_C(s)$	Partial fractions	$v_C(t)$
$\delta(t)$	$\frac{1}{1 + s\tau}$	$\frac{1/\tau}{s + 1/\tau}$	$\frac{1}{\tau} e^{-t/\tau}$
$u(t)$	$\frac{1}{s(1 + s\tau)}$	$\frac{1}{s} - \frac{1}{s + 1/\tau}$	$1 - e^{-t/\tau}$
$r(t)$	$\frac{1}{s^2(1 + s\tau)}$	$\frac{1}{s^2} - \frac{\tau}{s} + \frac{\tau}{s + 1/\tau}$	$t - \tau + \tau e^{-t/\tau}$



All three checked by simulating the circuit: at $t = 1 \text{ ms}$ and 3 ms the simulated and formula values agree.

Read the ramp result: after the transient dies, $v_C = t - \tau$. The output follows the input with a **constant lag of $\tau = 1 \text{ ms}$** that never goes away. That permanent error is why the ramp is called the toughest test.

CHECK 4

Differentiate the step response $1 - e^{-t/\tau}$. Which response do you get?

Exam form

Procedure for any first-order switching question. 1. $t = 0^-$: old circuit, DC steady state ($L = \text{short}$, $C = \text{open}$). Find $i_L(0^-)$ or $v_C(0^-)$. 2. $t = 0^+$: new circuit. $L \rightarrow \text{current source } i_L(0^-)$; $C \rightarrow \text{voltage source } v_C(0^-)$. Find the asked quantity. 3. $t = \infty$: new circuit, DC steady state ($L = \text{short}$, $C = \text{open}$). 4. $\tau = L/R_{Th}$ or $R_{Th}C$, with R_{Th} seen by the L or C . 5. $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$.

Laplace route. (1) Find $i_L(0^-)$, $v_C(0^-)$. (2) Transform: sources $\rightarrow F(s)$, $R \rightarrow R$, $L \rightarrow sL$ (+ source $L i_L(0^-)$), $C \rightarrow 1/sC$ (+ source $v_C(0^-)/s$). (3) KVL/KCL in s , solve for the unknown. (4) Partial fractions, inverse transform.

Test signals. $\delta(t) \leftrightarrow 1$, $u(t) \leftrightarrow 1/s$, $r(t) = t \cdot u(t) \leftrightarrow 1/s^2$; $\delta = du/dt$, $u = dr/dt$. RC ($\tau = RC$): impulse $(1/\tau)e^{-t/\tau}$, step $1 - e^{-t/\tau}$, ramp $t - \tau + \tau e^{-t/\tau}$.

Facts to state. Inductor current and capacitor voltage cannot change instantaneously ($v_C(0^-) = v_C(0^+)$, $i_L(0^-) = i_L(0^+)$). Settling $\approx 5\tau$ (99.3 %). 63.2 % of the change happens in one τ .

Traps

- Using the ∞ circuit at 0^+ (treating L as a short immediately).
- Calling $i_L(0^+) = 0$ when the inductor was carrying current at 0^- : it keeps its 0^- value, whatever it was.
- $\tau = L/R$ with the wrong R : it must be the Thévenin resistance the L sees.
- Using the first-order formula on an RLC circuit.
- Laplace: forgetting the initial-condition source, or writing a DC source as V instead of V/s .
- Using the final-value theorem on a response that oscillates forever.

Self-test

1. What does a capacitor look like at 0^+ if it was uncharged? And at ∞ ?
2. Why is $\tau = L/R$ and not LR ? Check with units: $H/\Omega = ?$
3. RL circuit, $i(0^+) = 2 \text{ A}$, $i(\infty) = 0$, $\tau = 5 \text{ ms}$. Write $i(t)$. What is i at 10 ms?
4. After how many time constants is a transient 99 % done?
5. Why is there no transient in a purely resistive circuit?
6. Give the Laplace transform of $\delta(t)$, $u(t)$, $r(t)$, and one everyday example of each.
7. Find $I(s)$ and $i(t)$ for an RL circuit ($R = 1 \text{ k}\Omega$, $L = 1 \text{ H}$) with $i(0^-) = 5 \text{ mA}$ and no source.

Answers

Check 1. $v_C(0^+) = 4 \text{ V}$, so the resistor sees $10 - 4 = 6 \text{ V}$: $i(0^+) = \mathbf{6 \text{ mA}}$. At ∞ the C is open: $i(\infty) = \mathbf{0}$ ($v_C(\infty) = 10 \text{ V}$).

Check 2. $\tau = 2 \text{ k}\Omega \times 5 \text{ }\mu\text{F} = \mathbf{10 \text{ ms}}$; settled after about $5\tau = \mathbf{50 \text{ ms}}$.

Check 3. $I(s) = \frac{10/s}{2000 + 1/(10^{-6}s)} = \frac{0.005}{s + 500} \Rightarrow i(t) = \mathbf{5e^{-500t} \text{ mA}}$ ($\tau = 2 \text{ ms}$; $i(0^+) = 10 \text{ V}/2 \text{ k}\Omega \checkmark$).

Check 4. $\frac{1}{\tau}e^{-t/\tau}$: the **impulse response**, as $\delta = du/dt$ predicts.

Self-test.

1. A **short** at 0^+ ; an **open** at ∞ .
2. $H/\Omega = (V \cdot s/A)/(V/A) = \mathbf{s}$: L/R is a time. (LR has units $\Omega^2 \cdot s$, not a time.)
3. $i(t) = \mathbf{2e^{-t/5 \text{ ms}} \text{ A}}$; at 10 ms , $i = 2e^{-2} = \mathbf{0.27 \text{ A}}$.
4. About $\mathbf{5\tau}$ (99.3 %); 4.6τ gives exactly 99 %.
5. A resistor stores no energy, so nothing needs time to build up or run down: every quantity jumps straight to its final value.
6. $\mathbf{1, 1/s, 1/s^2}$. Lighter spark (impulse), switch (step), accelerator (ramp).
7. $I(s) = \frac{L i(0^-)}{R + sL} = \frac{0.005}{s + 1000} \Rightarrow i(t) = \mathbf{5e^{-1000t} \text{ mA}}$ ($\tau = 1 \text{ ms}$).

09 — Second-order circuits: series RLC

MTE syllabus item 7 (under-, over-, critically damped, undamped) · not yet in the class notes: the question is a textbook problem · needs 08 ($0^-/0^+/\infty$, Laplace)

Map

1 Why RLC is second order

→ 2 The characteristic equation

→ 3 α and ω_0 decide everything

→

4 The four cases

→

5 Using the initial conditions

The question this file answers

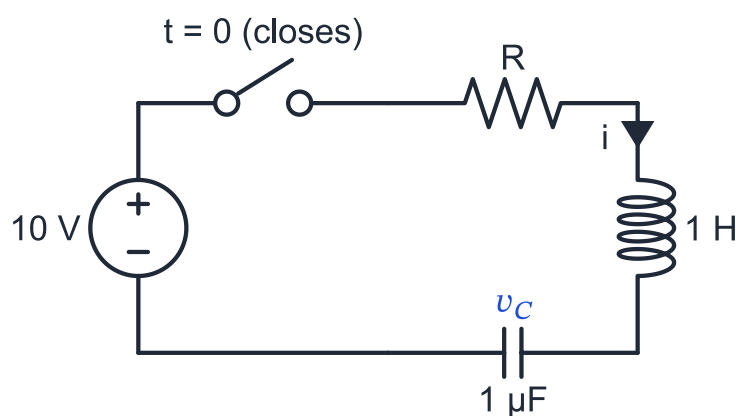
- **textbook** 10 V is switched onto a series R, 1 H, 1 μF circuit at $t = 0$ (everything at rest). Find $v_C(t)$ for $R = 2.5 \text{ k}\Omega$, 2 $\text{k}\Omega$, 1.2 $\text{k}\Omega$ and 0. Classify each.

The class notes (set 2 p.18) already mark the boundary: the one first-order formula is "not for RLC and LC circuits". This file is what replaces it. Facts from the circuits knowledge base (Hayt; Van Valkenburg; Alexander & Sadiku ch. 8); every curve below was checked by simulating the circuit.

Build

TEXTBOOK

Find $v_C(t)$ after the switch closes, for $R = 2.5 \text{ k}\Omega$, 2 $\text{k}\Omega$, 1.2 $\text{k}\Omega$ and 0.



$L = 1 \text{ H}$, $C = 1 \mu\text{F}$, source 10 V, switch closes at $t = 0$. $i(0^-) = 0$, $v_C(0^-) = 0$.

GUESS FIRST

Make R very large, then zero. What does v_C do in each case? Can it ever go **above** 10 V?

1 · Why it is second order

The circuit has **two** energy stores: the inductor (current) and the capacitor (voltage). Energy can pass back and forth between them, like a mass on a spring. The resistor is the friction.

KVL round the loop, with $i = C \frac{dv_C}{dt}$:

$$L C \frac{d^2 v_C}{dt^2} + R C \frac{dv_C}{dt} + v_C = V$$

A second derivative: one time constant is no longer enough.

2 • The characteristic equation

By Laplace (08 §6), with everything at rest: $I(s) = \frac{V/s}{R + sL + 1/sC}$. Multiply top and bottom by s/L :

$$I(s) = \frac{V/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

The denominator set to zero is the **characteristic equation**. Its roots (the poles) decide the shape of the response:

$$s^2 + 2\alpha s + \omega_0^2 = 0 \quad s = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

3 • Two numbers decide everything

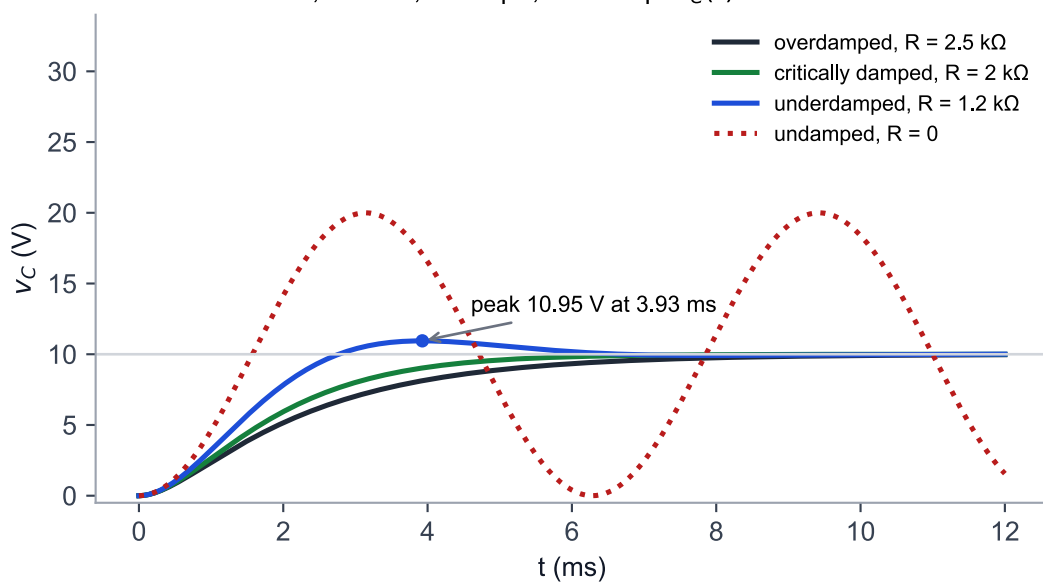
Symbol	Name	Series RLC	Here ($L = 1 \text{ H}$, $C = 1 \text{ } \mu\text{F}$)
α	damping coefficient (neper frequency)	$\frac{R}{2L}$	$\frac{R}{2}$ per second
ω_0	undamped natural frequency	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{(10^{-6})}} = \mathbf{1000 \text{ rad/s}}$
ζ	damping ratio	$\frac{\alpha}{\omega_0}$	

Compare α with ω_0 . The **critical resistance** is where $\alpha = \omega_0$: $R_c = 2\sqrt{\frac{L}{C}} = 2\sqrt{(10^6)} = \mathbf{2 \text{ k}\Omega}$.

4 · The four cases

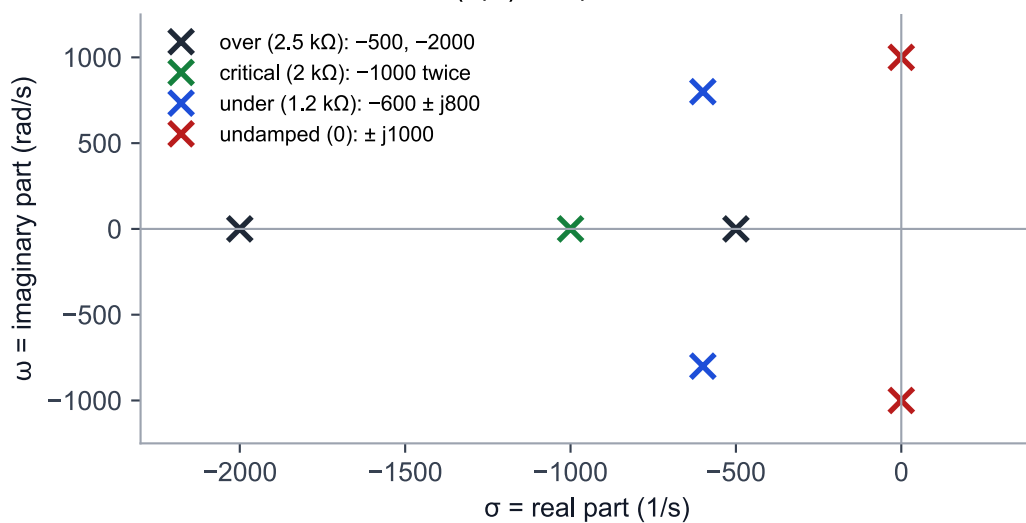
R	α	Roots	Case	$v_C(t)$ (V)
2.5 k Ω	$1250 > \omega_0$	-500, -2000 (real, different)	overdamped	$10 - \frac{40}{3}e^{-500t} + \frac{10}{3}e^{-2000t}$
2 k Ω	$1000 = \omega_0$	-1000, twice (real, equal)	critically damped	$10 - 10(1 + 1000t)e^{-1000t}$
1.2 k Ω	$600 < \omega_0$	$-600 \pm j800$ (complex)	underdamped	$10 - e^{-600t}(10 \cos 800t + 7.5 \sin 800t)$
0	0	$\pm j1000$ (imaginary)	undamped	$10(1 - \cos 1000t)$

Series RLC, L = 1 H, C = 1 μ F, 10 V step: $v_C(t)$ for four values of R



Each formula was checked against a time simulation of the circuit at $t = 1, 2, 3.93$ and 5 ms; they agree to 6 digits.

Roots of $s^2 + (R/L)s + 1/LC = 0$ for the four cases



Where the roots sit. On the real axis: no oscillation. Off it: oscillation at the imaginary part. The further left, the faster the decay.

What each case looks like:

- **Overdamped** ($\alpha > \omega_0$, $\zeta > 1$): two decaying exponentials; **slow**, no overshoot. The slow root (-500 , i.e. $\tau = 2$ ms) dominates.
- **Critically damped** ($\alpha = \omega_0$, $\zeta = 1$): the form $(A + Bt)e^{-\alpha t}$; the **fastest rise with no overshoot**.
- **Underdamped** ($\alpha < \omega_0$, $\zeta < 1$): a decaying oscillation $e^{-\alpha t} \cos(\omega_d t + \phi)$, at the **damped frequency** $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)} = \sqrt{(1000^2 - 600^2)} = 800$ rad/s. It **overshoots**: peak **10.95 V** at $t = \pi/\omega_d = 3.93$ ms.
- **Undamped** ($R = 0$, $\alpha = 0$): the energy swings between L and C forever at ω_0 ; v_C goes up to **20 V**, twice the source.

The Guess answer: large $R \rightarrow$ slow creep up to 10 V; $R = 0 \rightarrow$ endless oscillation between 0 and 20 V. So yes, v_C **can** exceed the source voltage, whenever the circuit is underdamped.

CHECK 1

Series RLC with $L = 10$ mH, $C = 1$ μ F. (a) Find ω_0 and the critical R . (b) Classify $R = 100 \Omega$ and $R = 500 \Omega$.

5 • Using the initial conditions

The table's constants come from the two initial conditions (08: what can't jump):

- $v_C(0^+) = v_C(0^-) = 0$ (capacitor voltage cannot jump).
- $i(0^+) = i(0^-) = 0$ (inductor current cannot jump), and since $i = C \frac{dv_C}{dt}$, this gives $\frac{dv_C}{dt}(0^+) = 0$.
- Final value: at ∞ the capacitor is open, the inductor a short, so $v_C(\infty) = 10$ V.

Overdamped, worked: $v_C = 10 + A_1 e^{-500t} + A_2 e^{-2000t}$.

$$v_C(0) = 0: A_1 + A_2 = -10 \qquad \frac{dv_C}{dt}(0) = 0: -500A_1 - 2000A_2 = 0$$

The second gives $A_1 = -4A_2$, so $-3A_2 = -10$: $A_2 = \frac{10}{3}$, $A_1 = -\frac{40}{3}$, as in the table.

Underdamped current, the other quantity asked most often: $i(t) = \frac{V}{\omega_d L} e^{-\alpha t} \sin \omega_d t = 12.5 e^{-600t} \sin 800t$ mA (simulation at 1 ms: 4.92 mA $\checkmark$).

CHECK 2

Why must the current in the undamped case be $i(t) = 10 \sin(1000t)$ mA? Check it with $i = C \cdot dv_C/dt$.

Exam form

Series RLC: characteristic equation $s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$; $\alpha = \frac{R}{2L}$, $\omega_0 = \frac{1}{\sqrt{LC}}$, $\zeta = \alpha/\omega_0$, roots $s = -\alpha \pm \sqrt{(\alpha^2 - \omega_0^2)}$.

Condition	Case	Form of the natural response
$\alpha > \omega_0$ ($R > 2\sqrt{L/C}$)	overdamped	$A_1 e^{s_1 t} + A_2 e^{s_2 t}$
$\alpha = \omega_0$ ($R = 2\sqrt{L/C}$)	critically damped	$(A_1 + A_2 t) e^{-\alpha t}$
$\alpha < \omega_0$	underdamped	$e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$, $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$
$\alpha = 0$ ($R = 0$)	undamped	$A_1 \cos \omega_0 t + A_2 \sin \omega_0 t$

Step response = final value + natural response. Constants from $v_C(0^+)$ and $i_L(0^+)$ (i.e. $dv_C/dt(0^+) = i_L(0^+)/C$).

Parallel RLC (for recognition): same ω_0 , but $\alpha = \frac{1}{2RC}$, so a **small** R gives heavy damping, the opposite of series.

Traps

- Using $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$ on an RLC circuit (set 2 p.18 warns against it).
- Writing $\alpha = R/L$ instead of $R/2L$.
- Oscillating at ω_0 in the underdamped case: the oscillation is at ω_d , which is smaller.
- Forgetting the second initial condition ($dv_C/dt(0^+)$ from the inductor current).
- Parallel RLC: using the series α .

Self-test

1. Write the characteristic equation of a series RLC and define α and ω_0 .
2. $L = 0.1$ H, $C = 10$ μ F. Find the critical resistance.
3. For $R = 1.2$ k Ω in the question, what is the period of the ringing?
4. Which case settles fastest without overshoot?
5. Why can an LC circuit ($R = 0$) never reach a steady state?

Answers

Check 1. (a) $\omega_0 = 1/\sqrt{(10^{-2} \times 10^{-6})} = \mathbf{10\,000\ rad/s}$; $R_C = 2\sqrt{(10^{-2}/10^{-6})} = \mathbf{200\ \Omega}$. (b) $100\ \Omega$: underdamped ($\alpha = 5000 < 10\,000$). $500\ \Omega$: overdamped ($\alpha = 25\,000 > 10\,000$).

Check 2. $i = C\,dv_C/dt = 10^{-6} \times 10 \times 1000 \sin 1000t = 0.01 \sin 1000t\ \text{A} = \mathbf{10\ sin(1000t)\ mA}$. It starts at 0, as the inductor requires.

Self-test.

1. $s^2 + (R/L)s + 1/LC = 0$; $\alpha = R/2L$, $\omega_0 = 1/\sqrt{LC}$.

2. $R_C = 2\sqrt{(0.1/10^{-5})} = 2 \times 100 = \mathbf{200\ \Omega}$.

3. $T = 2\pi/\omega_d = 2\pi/800 = \mathbf{7.85\ ms}$.

4. **Critically damped.**

5. No resistor to turn the energy into heat: it keeps passing between L and C, so the roots sit on the $j\omega$ axis and never decay.

10 — Mock paper (MTE syllabus)

Covers the professor's MTE syllabus (10/9/26), Parts I–III · all 14 of the professor's board questions + textbook problems for syllabus items the class has not yet worked · marks unknown (no hand-out or past paper yet) · attempt closed-book, then mark with the answers

HOW TO USE THIS

Do it in one sitting without opening 01–09. Write every step: the working earns marks, not just the final number. Mark each item clean / with struggle / wrong, and schedule it as in 00. Items tagged **textbook** are not the professor's; everything else is quoted from the board.

Section A — Theory

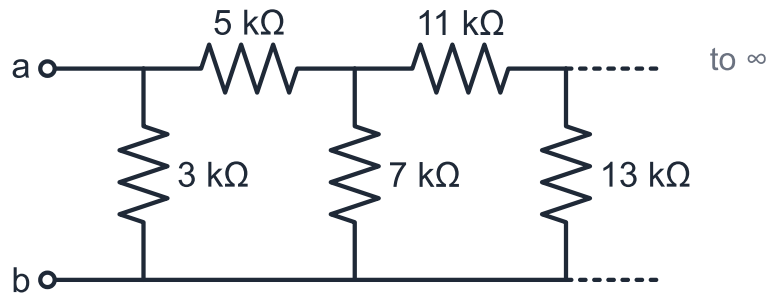
Prompts on the syllabus topics. No past paper has been seen yet, so the *wording* of these is not the professor's.

1. Classify circuit elements. Distinguish active and passive elements, with examples. Draw the symbols of the four dependent sources and name a device for each.
2. Distinguish lumped and distributed parameters.
3. Derive $v = L \, di/dt$ from $\psi = N\phi$. Explain why an inductor does not allow a sudden change of current.
4. State KCL and KVL. State the passive sign convention for power.
5. Write the procedures for nodal and mesh analysis. When is each preferred?
6. State the superposition theorem. List its limitations. How is it applied when sources have different frequencies?
7. Write the procedure for Thévenin's theorem. How is R_{Th} found when the network contains dependent sources?
8. State Norton's theorem. Give its procedure and draw the equivalent model.
9. Derive the condition for maximum power transfer to a resistive load. State the condition for a complex load.
10. Explain $t = 0^-$, 0^+ and ∞ . Write the general first-order response and the time constants of RL and RC circuits.
11. Define the unit impulse, step and ramp signals, with their Laplace transforms and one example each.
12. For a series RLC circuit, write the characteristic equation and classify the response by damping.

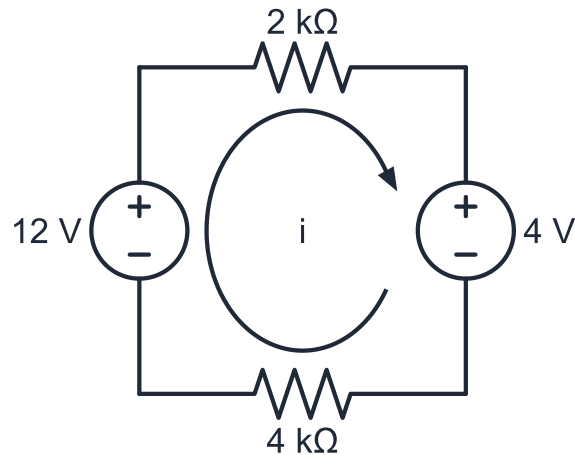
Section B — Numericals

Part I · elements, laws, analysis

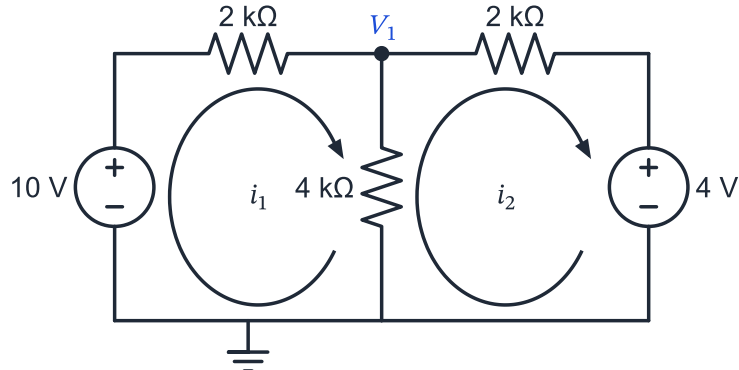
- B1** **Class p.4** 15 V is applied across 3 k Ω and 6 k Ω in series. Find V_1 and V_2 .
- B2** **Class p.5** A 12 A current source feeds 3 k Ω , 6 k Ω and 4 k Ω in parallel. Find I_1 , I_2 , I_3 .
- B3** **Class p.5** $v = 10 \sin(2000t + 30^\circ)$ V across 3 k Ω and 7 k Ω in series. Find v_1 , v_2 .
- B4** **Class p.1** Find R_{eq} of the infinite ladder 3 k Ω shunt, 5 k Ω series, 7 k Ω shunt, 11 k Ω series, 13 k Ω shunt, ...



B5 [textbook](#) Find the loop current and the power of every element. Show that the powers balance.

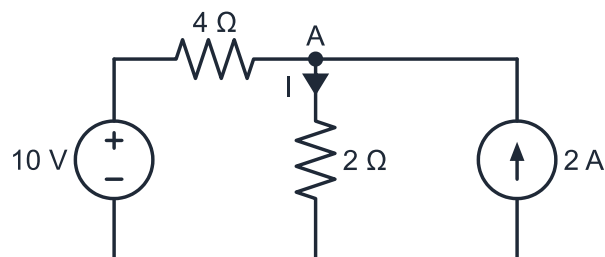


B6 [textbook](#) Find V_1 by nodal analysis and the mesh currents i_1 , i_2 by mesh analysis. Check that they agree.

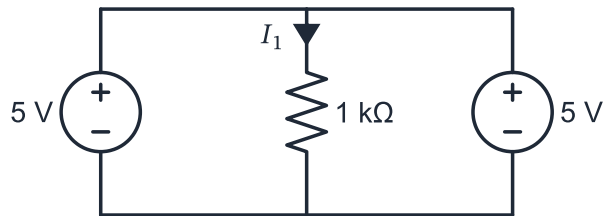


Part II · theorems (DC and AC)

B7 [Class p.7 · set 2 p.8](#) Calculate the current I using superposition. Verify by nodal analysis.

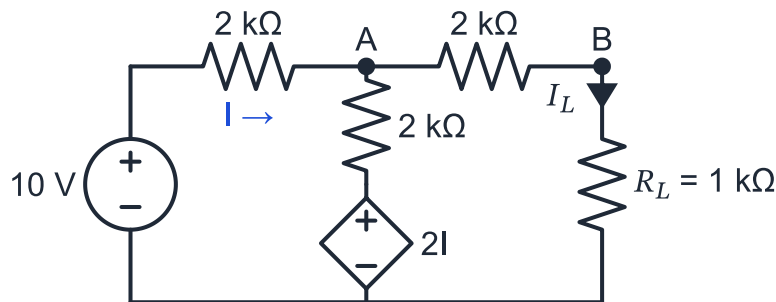


B8 [Class p.9 · set 2 p.9](#) Calculate the current I_1 using superposition. Comment on the result.



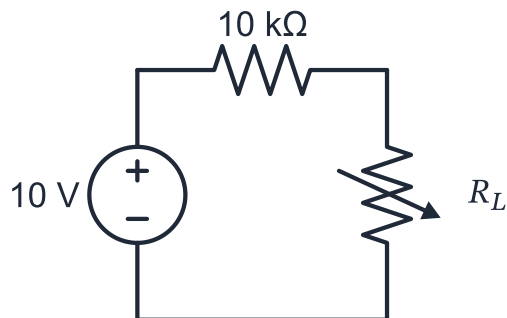
B9 [textbook](#) A 10 V DC source and a $10 \cos(1000t)$ V source in series drive $1 \text{ k}\Omega$ and 1 H . Find $i(t)$ in steady state.

B10 [Class p.11 · set 2 p.11](#) Determine the load current using Thévenin's theorem (take $2I$ as $2 \text{ k}\Omega \times I$, I in mA). Verify with Norton's theorem.



B11 [textbook](#) $10\angle 0^\circ \text{ V}$ at $\omega = 1000 \text{ rad/s}$, $1 \text{ k}\Omega$ in series, $1 \mu\text{F}$ across the output. Find the Thévenin and Norton equivalents, the load for maximum power, and that power.

B12–B13 [Class p.16–17 · set 2 p.14](#) For the circuit below: (a) the value of R_L for maximum power; (b) the total power delivered by the source; (c) the power consumed by R_{Th} ; (d) the efficiency; (e) the voltage across the load; (f) the relation between load power and total power, and between load voltage and source voltage.



B14 [Class p.17](#) 10 V source with $5 \text{ k}\Omega$: determine R for maximum power, the maximum power, and the efficiency.

Part III · transients

B15 [Class p.19–20](#) V , switch closing at $t = 0$, R . Find $I(0^-)$, $I(0^+)$, $I(\infty)$.

B16 [Class p.21](#) V , switch closing at $t = 0$, R and L in series. Find I_R , I_L at 0^- , 0^+ , ∞ . Then find $i(t)$ for $V = 10 \text{ V}$, $R = 1 \text{ k}\Omega$, $L = 1 \text{ mH}$, **by Laplace transform**.

B17 [Set 2 p.20–21](#) V , switch closing at $t = 0$, R and C in series (C uncharged). Find v_C , v_R , I at 0^- , 0^+ , ∞ .

B18 **Class p.22** 10 V, a switch opening at $t = 0$, $1\text{ k}\Omega$ and 1 mH in series. Find $I_L(0^-)$ and $I_L(0^+)$. What happens to the inductor's current after the switch opens?

B19 **textbook** An RC circuit ($R = 1\text{ k}\Omega$, $C = 1\text{ }\mu\text{F}$, output v_C) is driven by a unit step, then by a unit ramp. Find $v_C(t)$ for each by Laplace transform.

B20 **textbook** 10 V is switched onto a series R, 1 H , $1\text{ }\mu\text{F}$ circuit (at rest). Classify the response for $R = 2.5\text{ k}\Omega$, $2\text{ k}\Omega$, $1.2\text{ k}\Omega$, 0 , and find $v_C(t)$ for $R = 2.5\text{ k}\Omega$.

Answers

Section A: model answers are the **Exam form** sections: 01 (A1–A3), 02–03 (A4), 04 (A5), 05 (A6), 06 (A7, A8), 07 (A9), 08 (A10, A11), 09 (A12).

	Answer	Where it's taught
B1	$V_1 = 5 \text{ V}, V_2 = 10 \text{ V}$	02 §1
B2	5.33 A, 2.67 A, 4 A	02 §2
B3	$v_1 = 3 \sin(2000t + 30^\circ) \text{ V}, v_2 = 7 \sin(2000t + 30^\circ) \text{ V}$	02 §3
B4	$\approx 2.318 \text{ k}\Omega$ (if the values continue as primes)	02 §5
B5	$i = 4/3 \text{ mA}$; 12 V source -16 mW (delivers), 4 V source $+5.33 \text{ mW}$, $2 \text{ k}\Omega + 3.56 \text{ mW}$, $4 \text{ k}\Omega + 7.11 \text{ mW}$; sum 0	03 §2–3
B6	$V_1 = 5.6 \text{ V}$; $i_1 = 2.2 \text{ mA}$, $i_2 = 0.8 \text{ mA}$; $4 \text{ k}\Omega$ carries $1.4 \text{ mA} = 5.6 \text{ V}/4 \text{ k}\Omega \checkmark$	04 §1, §3
B7	$I_1 = 5/3 \text{ A}$, $I_2 = 4/3 \text{ A}$, $I = 3 \text{ A}$ (resistors in Ω); nodal $V_A = 6 \text{ V}$	05 §1–3, 04 §2
B8	By the rule each case gives 0 (the shorted idle source also shorts the $1 \text{ k}\Omega$), so $I = 0 \text{ A}$; true value 5 mA . Limitation: superposition fails for parallel ideal voltage sources	05 §5
B9	$i(t) = 10 + 7.07 \cos(1000t - 45^\circ) \text{ mA}$	05 §6
B10	$V_{Th} = 20/3 \text{ V}$, $R_{Th} = 8/3 \text{ k}\Omega$, $I_L = 20/11 \approx 1.82 \text{ mA}$; Norton $I_N = 2.5 \text{ mA}$, same I_L . (With $2I$ read as 2Ω : 5.0025 V , $3.0 \text{ k}\Omega$, 1.25 mA ; see 06 §2 Note)	06 §2–5
B11	$V_{Th} = 7.07 \angle -45^\circ \text{ V}$, $Z_{Th} = 500 - j500 \Omega$, $I_N = 10 \angle 0^\circ \text{ mA}$; $Z_L = 500 + j500 \Omega$, $P_{max} = 12.5 \text{ mW}$	06 §6, 07 §4
B12–B13	(a) $10 \text{ k}\Omega$ (b) 5 mW (c) 2.5 mW (d) 50% (e) 5 V (f) $P_L = P_s/2$, $V_L = V_s/2$	07 §2–3
B14	$R = 5 \text{ k}\Omega$, $P_{max} = 5 \text{ mW}$, $\eta = 50 \%$	07 Check 1
B15	0, V/R , V/R	08 §1
B16	0, 0, V/R for both; $I(s) = 10^4/(s(s + 10^6)) \Rightarrow i(t) = 10(1 - e^{-t/1 \mu s}) \text{ mA}$	08 §2, §6
B17	0^- : all 0. 0^+ : $v_C = 0$, $v_R = V$, $I = V/R$. ∞ : $v_C = V$, $v_R = 0$, $I = 0$	08 §2
B18	$I_L(0^-) = I_L(0^+) = 10 \text{ mA}$. With no path, the ideal circuit has no solution (voltage spike, arc). If the R–L loop closes on itself: $i(t) = 10e^{-t/1 \mu s} \text{ mA}$	08 §5
B19	$\tau = 1 \text{ ms}$. Step: $1 - e^{-t/\tau} \text{ V}$. Ramp: $t - \tau + \tau e^{-t/\tau}$ (lags the input by τ)	08 §7
B20	$\omega_0 = 1000 \text{ rad/s}$, $R_c = 2 \text{ k}\Omega$: over, critical, under, undamped. $R = 2.5 \text{ k}\Omega$: $v_C = 10 - (40/3)e^{-500t} + (10/3)e^{-2000t} \text{ V}$	09 §4–5